Implementation Guidelines for the Industrial Internet of Things

Mini Dissertation

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Implementation Guidelines for the Industrial Internet of Things

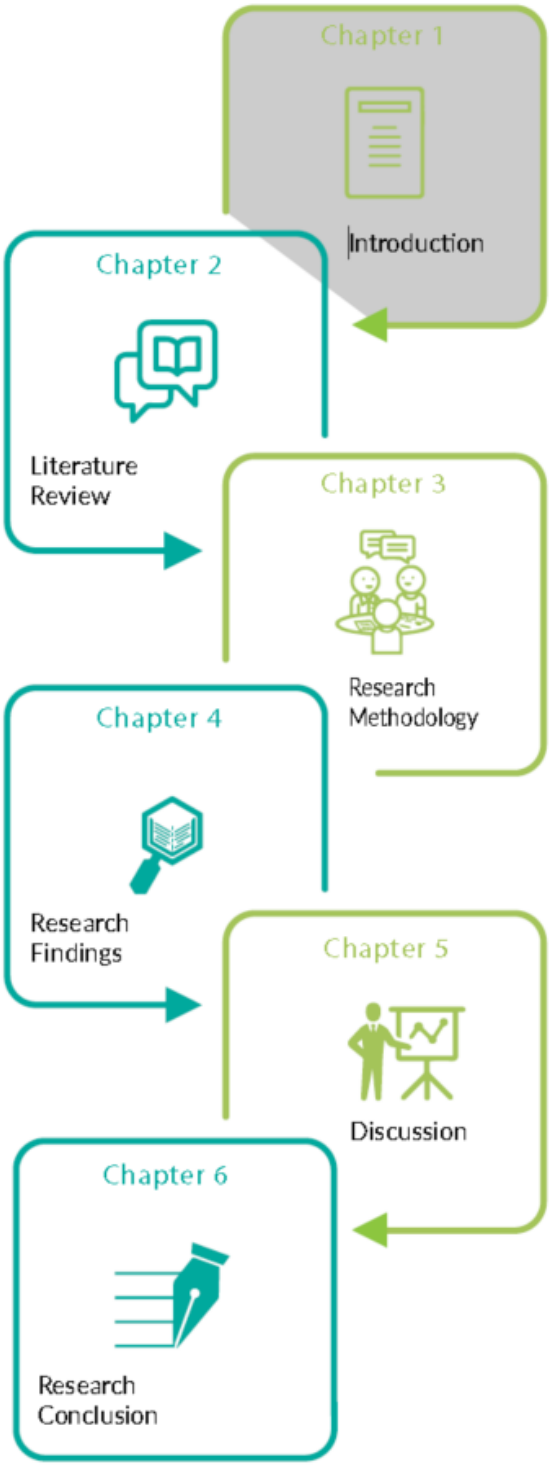
ABSTRACT

The Industrial Internet of Things is a new concept that has the potential to add value to any industrial organisation that wants to embark on the implementation thereof. Within this study, the researcher explored a set of guidelines that industrial organisations can use for the successful implementation of the technology.

Keywords: cloud services, Industrial Internet of Things, Machine-to-Machine communication, Supervisory Control and Data Acquisition

ABBREVIATIONS

IIoT	Industrial Internet of Things
M2M	Machine-to-Machine communication
OSI	Open Systems Interconnection
SCADA	Supervisory Control and Data Acquisition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CHAPTER 1: INTRODUCTION

In 1926 Nicola Tesla (1926), in an interview with Colliers magazine, said the following: "When wireless is perfectly applied the whole earth will be converted into a huge brain, which in fact it is, all things being particles of a real and rhythmic whole … the instruments through which we shall be able to do this will be amazingly simple compared with our present telephone. A man will be able to carry one in his vest pocket."

With this statement in 1926, visionary Tesla described the same principles applied to the internet of things (IoT). Muntjir, Rahul and Alhumyani (2017) describe the IoT as the link between the physical world on the one hand and the cyberworld on the other with the help of things or objects that possess sensing abilities to transmit the measured results via a network to achieve a purpose. There is an ever-increasing number of objects, from home appliances to smartphones, daily connecting to the internet or internet-like structure

The IoT has the ability to change the way that organisations and individually more significant industries perform in terms of safety and production. Safety and production improvements would have a positive impact on any large industrial sector. This study aims to provide guidelines for the implementation of the IoT in industrial operations, for instance, a mining operation. The guidelines will address the existing gap in industrial operations in terms of Industrial Internet of Things (IIoT) implementations. This gap exists since there is a need for

the benefits of the IIoT, but not many implementations have been completed. These guidelines, if correctly applied, will result in a successfully implemented IoT platform that would realise the value that IoT can unlock. The starting point for IoT-specific decisions or prospective implementers is guidelines. These guidelines will assist in determining whether or not an organisation will benefit from such an implementation and also provide the correct course of action when implementing IoT solutions.

1.1 BACKGROUND INFORMATION

Mining, as an industrial representor, plays an essential economic role in South Africa. The Chamber of Mines (2017) showed that the employee earnings within the mining sector amounted to R120 billion, with mining contributing R304.4 billion to the gross domestic product for the year 2016. The mining sector created 457,698 jobs and supported approximately 4.5 million dependants.

Statistics South Africa paints a different picture – one that shows the rapid decline in mining and mining-related activities. “Historical values of the gold index show the extent of how production has fallen. In January 1980, the index was 359.0, while the volume of gold produced was far lower in January 2015, resulting in the low index of 48.4. In other words, South Africa produced 87% less gold in January 2015 compared to the same month in 1980” (Statistics South Africa, 2015).

South Africa’s gold sector, which was once the world-leading gold producer, needs to increase its mining production efficiency to avoid vanishing entirely by the year 2020, especially after being rated sixth in the world (Major, 2015). The statistical data from Statistics South Africa confirm Major’s claim.

Major (2015) was addressing mining representatives at the Johannesburg Stock Exchange Power Hour when he said that “gold mining has lost a million people in 30 years”. Major (2015) explained, “People say our gold mines are closing because of lower grade ore, but we know better than that. It is not about grades; it is about efficiencies and cost. We are

taking out as many ounces of gold per employee now as we were in 1907 when we were using picks and shovels."

In early 2015 at the Mining Indaba, the CEO of Anglo American Platinum, Chris Griffith, emphasised the need to modernise the mining sector if people want to keep on mining and (Statistics South Africa, 2015) to stay a leading employment sector. Chris Griffith followed up with this explanation: "For us at Amplats, that involves moving away from the past, conventional, labour-intensive underground mining, with its high demands for people, energy and expensive infrastructure to a modern way of mining."

Further to this, the research of O'Conner (2017) focuses on the challenges that the South African mining industry faces, including:

- A reduction of water and energy resources
- Employee health and safety
- Improved production by the use of automation
- Reduction in waste

Merry (2017) states that an IoT implementation could address these challenges as mentioned by O'Conner as follows. To reduce waste, the mining progression could be tracked from where the mining groundworks start by collecting data with the help of IoT. This data would then be used to manage the building of the mine. Merry (2017) further explains that the advancement of safety aspects could be achieved with the help of IIoT by monitoring dangerous slopes or mining shafts in real-time. In 2015, a total of 3116 injuries were reported, according to the Department of Mineral Resources of South Africa. Any reduction in injuries would have a direct favourable effect for the industry. Merry (2017) says that production can be increased with the help of IoT by a higher level of automation. This higher level of automation with the help of IoT would enable current stand-alone equipment to work together. Further predictive maintenance could be complemented by the remote monitoring of equipment, and this could have a reduction in the cost of energy (Merry, 2017).

In the past, mining companies installed sensors only at critical points that were needed to control the operations. The reason for installing sensors at critical points only was because of the cost of these sensors. The cost has drastically reduced due to an increased demand in sensors. The production costs of sensors were consequently drastically lowered, making it highly possible to add sensors to more measuring points (Holdowsky, J., Mahto, M., Raynor, E. R. & Cotteleer, M, 2019). With the increase in measuring points by adding more sensors, the amount of data collected could tell a lot about the state of the asset, which is not limited to the temperature, vibrations and energy of the assets to which it is connected. By adding intelligence to these sensors, different sensors could communicate with each other, thereby creating intellectual assets. Connecting all the sensors or “things” through the internet or data communications infrastructure, an intelligent network of machines or things was created that could communicate with each other and feed data to analytics engines to identify patterns of deviation in assets. These analytics and applications that interpret the data could typically alert people to act upon.

To lower costs, improve processes and safety and increase efficiency, one could collect and analyse data collected from intelligent assets. Intelligent connected devices could identify problems specific to their assets with the help of the analytics engine’s machine-specific problems and could potentially help to reduce downtime by troubleshooting, remote diagnostics and condition monitoring capabilities. This, in turn, would improve performance. Gorbach and Polsonetti (2015) mention that IoT would support predictive maintenance approaches in the sense that it would reduce unplanned downtime with an improvement of maintenance productivity and effectiveness and would enable operation of assets in the most optimised manner. The ability to remotely access the data from these operational assets would lead to further incremental business value (Gorbach & Polsonetti, 2015). If remote monitoring were used, it could lead to reduced downtime by monitoring sensor data such as temperature and vibration for predictive maintenance. Gorbach and Polsonetti (2015) use the example of service personnel that could identify specific problems and perform configuration changes remotely, thereby saving them a trip to the plant.

1.2 PROBLEM STATEMENT

Sensor prices have had a steady decline in their pricing, therefore becoming more affordable (Raynor & Cotteleer, 2015). Parallel to these advancements in big data handling, artificial intelligence and machine learning, the IIoT could only recently be implemented in industrial sectors because of the maturity and affordability. The problem due to the hardware that only recently became affordable and the development of the analytic tools on an application layer (Holdowsky, et al., 2019)was that only a few IIoT implementations had been completed in the industrial environment. Further to this, literature or guidelines that could assist companies in implementing an industrial IIoT solution are few.

If relevant literature or guidelines in the implementation of IIoT existed on this topic, it would enable more industries to implement and leverage the advantages of IIoT in a shorter period too. The benefit of clear guidelines in the industrial sector would mean that implementations could realise some objectives of improvement on the safety and efficiency of an IIoT solution faster. Since there are no guidelines, it creates a gap in the literature and motivates this research.

1.3 PURPOSE OF THE STUDY

The purpose of this study was to define and discover implementation guidelines for IoT that apply to an industrial operation. This would direct any prospective IoT implementations within the industrial sectors and would ensure a higher positive outcome in terms of implementation within a specific industry. IoT in the industrial space is referred to as the IIoT. For the discovery of guidelines, it is necessary to review previous IIoT implementations within the industries and interview subject matter experts in the domain. This would include an investigation on the various components of the IIoT. These components include the complete architecture of the IIoT and the conditions that previous implementers deem essential for the success of the IIoT. A further investigation will be made into the applications that IIoT will feed information to, and the value that it could deliver.

1.4 RESEARCH QUESTIONS

Main research question: What is a list of guidelines that can be used for IIoT implementations?

Objective: Industrial companies could find it challenging to navigate through all the IIoT information to understand the guidelines to follow to implement an IIoT; in the planning phase, organisations can decide if they want to pursue an IIoT implementation.

Sub-question to the main questions:

- What guidelines are pertinent in the existing literature?
  The objective is to obtain implementation guidelines from existing literature.
- What are the critical considerations for any organisation planning to implement the IIoT to gain the advantages?
  Objective: If the organisation understands the building blocks that are needed for successful implementation, it could prevent unnecessary rework or implementation mistakes previously made by other installers and offer the opportunity to learn from these mistakes.
- What are the lessons learned from other implementations in the industrial organisations that could assist new implementations?
  Objective: Limited academic guidelines are available that are backed by the practical implementation within the industrial sphere.

1.5 ASSUMPTIONS/LIMITATIONS

The researcher trusted that the selected interviewees would provide trusted information. This could be achieved by focusing on interviewees from IIoT solution providers involved in solutions for industrial firms. The researcher further assumed that he would be able to gather the necessary data from the expert responders and that they would be available. This research was conducted within the industrial environment and the findings might not

necessarily be valid for all industries. Further studies could verify this. This study focused on IIoT implementation.

1.6 BRIEF CHAPTER OVERVIEW AND RESEARCH STRUCTURE

The research followed a structure consisting of six chapters. Chapter 1 offers the background to the study and introduces the reason this study was started. It covers the thesis statement and the critical questions that need answering within the study. Chapter 2 reviews existing literature in search of answers to the research questions. Chapter 3 suggests a research methodology in answering the research questions that were left unanswered in the literature study. The research methodology systematically explores the research concepts to arrive at a research method for data gathering. Chapter 4 reports on the research findings and focuses on the results. The data generated are presented and discussed in the same chapter. Chapter 5 discusses the research findings obtained from Chapter 4. Chapter 6 is the last chapter. It contains the recommendations and answers to the research question.

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CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

Within this literature review, both the topics of IoT and a subset of the broader IoT, namely IIoT, are explored because of the closed nature of the two topics. An overview of the IIoT architecture is also in the discussion. The architecture relevance is to understand the building blocks that need to be in place for a successful implementation of the IIoT footprint. This literature review explores the IIoT in broad terms to equip the reader with the background information on the topic. In Chapter 4, findings in the existing literature will contribute to addressing the research questions and the gap in the research topic. Expert interviews also formed part of extracting guidelines.

The IoT is an increasingly growing topic of interest and a regular discussion point in the information technology arena (Mitchell, n.d.). IoT could potentially change the traditional approach to the use of the internet. In the future, the goal of IoT is the unification of all “things” under a shared infrastructure, whereby network-connected sensors will be able to collect data from the surrounding environment. The collected data can then be shared across the internet to be processed and changed into meaningful information for various reasons (Mitchell, n.d.).

“IoT will transform the internet from a human-centric platform, where most applications are focused on human-to-human interaction and where most of the data are provided by humans, to a machine-intensive platform” (Pena, Rodriguez-Andina & Manic, 2017). Machine-to-machine (M2M) generated data, communication and interactions between “things” will initialise different sets of actions based on the surroundings and designed tasks. The operations and actions of these smart “things” will eventually be limitless; some examples include directing transport, where transportation machines will be able to adapt and be self-sustainable and even play a role in their disposal (Giusto, Lera, Morabito & Atzori, 2010).

IoT will have an impact on our daily lives. Data will be able to be collected, analysed and evaluated by numerous “things” such as light bulbs, water pipes and power outlets. With the availability of this collected data, people will be able to better incorporate technology with various institutions, for instance the balance of energy usage or the production performance of a company. This technology could have a positive value-adding benefit for big industrial organisations that consume large amounts of energy. According to Eskom (2008), South African mining industries as an industrial organisation are one of the biggest consumers of electricity at 15% of Eskom’s annual output. In mining, energy can be saved by applying efficiency solutions in the areas identified in Figure 1 below.

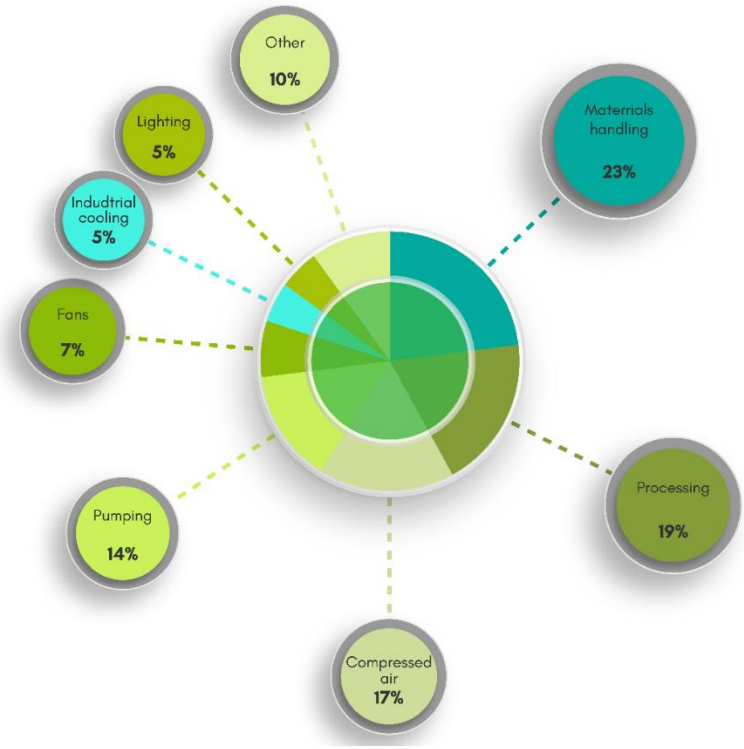

Figure 1: Energy consumption in the mining industry

The above energy example is one of the few possibilities that could present value by a successful IoT implementation within the mining sector. Within this literature review, emphasis will be placed upon the IIoT and IoT concepts to create an understanding of the components that it consists of and lessons from previous successful implementations in the industries through the systematic review of research journals, corporate white papers and books.

The Cambridge Dictionary defines guidelines as “information intended to advise people on how something should be done or what something should be” and implementation as “the act of putting a plan into action or of starting to use something” (Cambridge, 2019). When considering the implementation guidelines for an IIoT, the researcher focused on extracting the information that is presented in the literature as essential considerations in terms of the IIoT implementation.

Raynor and Cotteleer (2015) identify the different stages of information. The first is the creation stage, where sensors produce information about a physical occurrence. The second stage is the communication stage for the spread of data from one place to another and then aggregating or gathering the data from the different sources and times (Raynor & Cotteleer, 2015). The fourth stage is the analysis state that recognises relationships or patterns within the information that could help with predictions or descriptions for action. The last stage is the action stage of changing or maintaining a physical event or stage (Raynor & Cotteleer, 2015).

The literature study would provide input to implementation guidelines for an IIoT that apply to an industrial operation.

2.2 BACKGROUND TO IOT AND IIOT

2.2.1 WHAT IS THE IOT AND IIOT?

Muntjir et al. (2017) describe the IoT (2017) as the linkage between the physical world on the one hand and the cyberworld on the other with the help of things or objects that possess sensing abilities to transmit the measured results via a network to achieve a purpose. Within this definition, the basic building blocks that the IoT consists of are presented as the sensing objects used in the measurement of metrics such as flow, pressure, temperature position and vibration and the network that communicates the result to a higher decision-making engine. The same definition applies to the IIoT domain; the only difference is that the IIoT is mostly focused on industrial applications.

2.2.2 IOT AND IIOT DEFINITIONS

Further popular definitions of the IoT have different facets that are reliant upon the identity, the respective functionality and viewpoint of the institution that defined it. The emphasis is either on the sensors or physical hardware that collects the data, the transport layer or the analysis layer (Amazon, 2018). For instance, Amazon, which is one of the biggest online

shopping companies and analytics service providers in the cloud space, defines IoT as “a system of ubiquitous devices connecting the physical world to the cloud” (Amazon, 2018). The Working Group of The European Technology Platform on Smart Systems Integration (2008) defines IoT as a network of sensors that senses changes in data, for instance temperature and actuators that could react upon a motion to open a valve or garage door (things) to would work together and subsequently improve the quality of our lives with a reduction of mankind's footprint on the planet. Their perspective of a green future takes preference in the description. The viewpoint of the Working Group of The European Technology Platform on Smart Systems Integration (2008) focuses more on the improvement of lives in stark contrast to Amazon’s definition of connecting to the cloud, keeping in mind that Amazon sells cloud services. Amazon's main aim from this definition could be to make money and not as such the improvement of lives.

The focus of a group with a futuristic focus for the year 2020 describes IoT as “interconnected objects having an active role in what might be called the Future Internet”. The fundamental enabler is the identity knowledge of the object to know its properties and “itself” and the data that “itself” produces. The data pertain to its creation date, when to recycle itself, surrounding temperature and vibration or its intended use (The European Technology Platform, 2008). This definition values the Future Internet and focuses on this end to define the IoT.

The definition of IoT by Muntjir et al. (2017) is a more balanced approach that considers the lower and higher end of the IoT spectrum. In this regard, Muntjir et al. (2017) defines the IoT as connecting the physical (the things) and cyber worlds (the cloud), where an everyday object can sense, process and network characteristics of the object that it measures while allowing the object to communicate over the internet to accomplish some purpose. Madakam, Ramaswamy and Tripathi (2015) complemented this definition with: “An open and comprehensive network of intelligent objects that has the capacity to auto-organise, share information, data and resources, reacting and acting in the face of situations and changes in the environment.” This explanation supports the Muntjir et al. (2017) definition’s

"accomplishment of a purpose" by mentioning that the purpose is to react or act in environmental changes or situations.

All devices that contain a sensor, even smartphones and tablets, contribute to the increased network of devices that connect to the internet. In the industry, this includes things like production machinery, cars, jet engines and even oil drills. These devices collect data and exchange it over the internet to improve the process in which it is involved.

These elements combined maximise revenue by improving efficiency within the business. The operation of the Nestlé Nespresso Coffee Machine is an example that Frenzel used in his article. He says that "a successful 'internet of things' would take advantage of machine-to-machine communication that would include measuring and communicating between itself temperature, vibration and pressure for improved quality to advance customer satisfaction" (Frenzel, 2012).

The IIoT approach for improvement of performance and operations to add value is the usage of higher-level analytics, consolidation of data and leveraging experts that are physically remote from the plant. Predictive analytics can transform processes from manual/reactive to automatic/pro-active (Blanchette, 2016).

When using the IoT in mining and manufacturing (industrial operations), it is referred to as the IIoT (Karschia, 2015). This study focused on this aspect of IoT.

2.2.3 ALIASES OF IOT AND IIOT

The different names of the IoT are as diverse as the definitions of the IoT. The names are influenced by the viewpoints of the individuals or institutions that have a specific interest in technology. The objectives of the IoT are the same in a broad sense. It is also known as Web of Things, IIoT, Embedded Intelligence, Internet of Objects, Industry 4.0 and Connected Devices (Madakam et al., 2015).

Muntjir et al. (2017) in their work "An analysis of the internet of things" list the following perspectives and fields of IoT:

- Cyber-physical systems – Bringing virtual reality into the physical world through the integration of processes.
- Pervasive computing – When objects with internal processors are connected to a global network within a computer environment.
- Ubiquitous computing (calm technology) – When people will not even notice the technology that plays a role in their daily lives.
- Machine-to-machine interaction (M2M) – Within M2M communication, communication takes place between machines without human intervention.
- Human-computer interaction (HMI) – The interaction between humans and computers.
- Ambient intelligence – Made possible by the changing of the sensitivity and responsiveness of the everyday environment through technology.

2.2.4 IOT AND IIOT

The IoT is instrumental in delivering disruptive change to segments such as agriculture, healthcare, utilities and the government. The IIoT, a portion of the all-encompassing IoT, focuses on industries such as mining, oil and gas, manufacturing and utilities (Chan, 2017). Chan further explains the differences between IoT and IIoT under the following ten items (Table 1):

Table 1: Differences between the IoT Things and the IIoT

Item	IoT	IIoT	Explanation
Security	Low level	High level	IIoT has a higher level of security due to the critical nature if a high-volume manufacturing process were to be disturbed.
Reliability	IoT systems do not need a high level of reliability.	These conditions range from anything between high temperature, dust, pressure or vibration even in remote locations. IIoT solutions would need to withstand the same conditions.	Industrial systems could efficiently function, in harsh environments, anything between 20 and 30 years before the need for replacement.
Resilience	Failure of an IoT device would not have any catastrophic effects.	An essential aspect of industrial operations when systems cannot afford to fail because a breakdown in one part will stop operations. IIoT would need a high level of resilience.	If there is an occurrence of a loss in operational capability, a backup system would have taken over the process.
Programmability	IoT would be programmed for a specific function.	Industrial systems need to be frequently re-programmed to support new processes.	IIoT solutions that support these applications should be able to support the remote programming of these devices on-site or in the field.
Low latency	Latency is not important.	Low latency is needed because of the high-speed at which production takes place. An example is a food and beverage packaging plant.	In industrial organisations, deviations from the norm need to be detected almost in real-time to prevent any delays in worker safety, costs, quality

			of products or loss in revenue. IoT should, therefore, be able to support these low latency requirements.
Serviceability	IoT devices do not need a high level of serviceability. Other sensors could replace them.	IIoT devices should be easily maintainable, for instance easily swapped out.	Because of the expectation that industrial equipment should withstand harsh conditions for years, it should be easily maintainable. Part of maintainability entails the ease with which IIoT devices could be swapped out up to the ability to maintain these devices.
Automation	No need for a high level of automation in IoT solutions.	A need for a high level of automation in industrial environments. To achieve this, in some cases, intelligence is incorporated in the edge devices at the instrument level.	Some of these industrial systems have no human intervention. Any IIoT device that would need to function in the same manner should have the same abilities and also be programmable.
Precision and accuracy	Low level of accuracy needed.	High level of accuracy and precision required in industrial operations.	For example, in manufacturing environments, if there is an accuracy deviation, this could harm the whole manufactured batch.
Scalability	Not a high level of scalability needed.	Should be scalable into the thousands.	Scalability should take the scheduling, workflow integration, data collection and analysis into consideration because industrial networks support thousands of more of the same. More of the same in this regard could mean robots or controllers.
Interoperability	IoT consists mostly of new systems.	Industries have been in existence for over 20 years, and IIoT should consider the legacy systems.	IIoT needs to function with legacy systems. M2M and purpose-built manufacturing execution systems with different sets of data and different protocols.

2.2.5 TIMELINE OF IIOT

Figure 2 describes the progression of the IIoT since the creation of programmable logical controllers up to the present year.

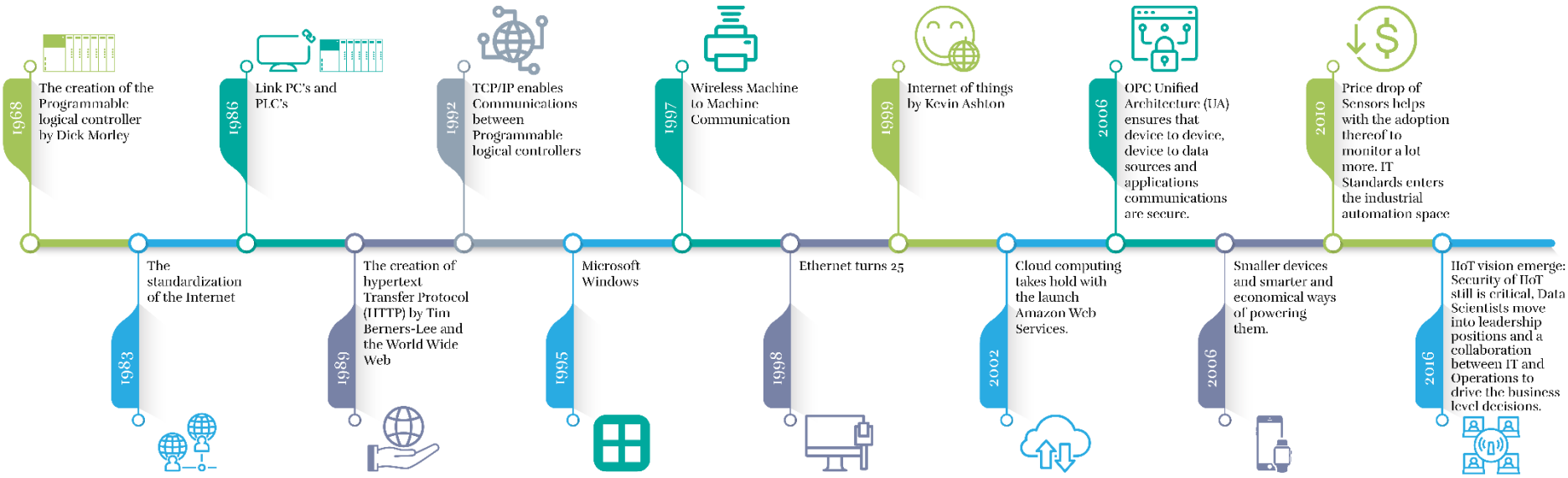

Figure 2: Timeline of the Industrial Internet of Things adapted from Machina Research (2015)

2.2.6 ADVANTAGES OF IIOT

Karschia (2015) mentions that there are four advantages of the IIoT connecting sensors to systems to analyse and improve performance, safety, reliability and energy efficiency. The reasons are, firstly, that data can be collected automatically from the sensors as it is less expensive, wireless and often battery-powered. The second reason is that information is gained by collecting data and strategically interpreting it through big data analytics and other techniques. These analytics engines are now in existence and can handle a great deal of data. Thirdly, with the availability of actionable information and with the help of the right persons, the findings from the information can be actioned for the benefits of the organisation. It does not matter if the resource is on- or offsite. Finally, the performance can be improved if corrective action would occur.

Merry (2017) from IBM lists the following five advantages. The IIoT is timesaving; for instance, before the initial digging of a mining pit takes place, data could be collected to ensure the proper management of the process. Secondly, safety can be improved by real-time monitoring and analysing slopes or unstable shafts. The advancement of automation could occur by having current stand-alone product equipment working together with vehicles after implementing the IoT. Furthermore, predictive maintenance with a fully integrated network that monitors every aspect of an organisation could detect wear-and-tear on vital pieces of equipment. Finally, there are energy and cost benefits through the reduction of energy expenditure and maintenance costs. Dua's (2016) article on the top five benefits of using IoT in mining is also in line with these named benefits. Morgan Stanley (2018) lists the benefits of IIoT in the graph in Figure 3 below, with improving operational efficiency as the leading initiative that could have a 47% benefit on the organisation.

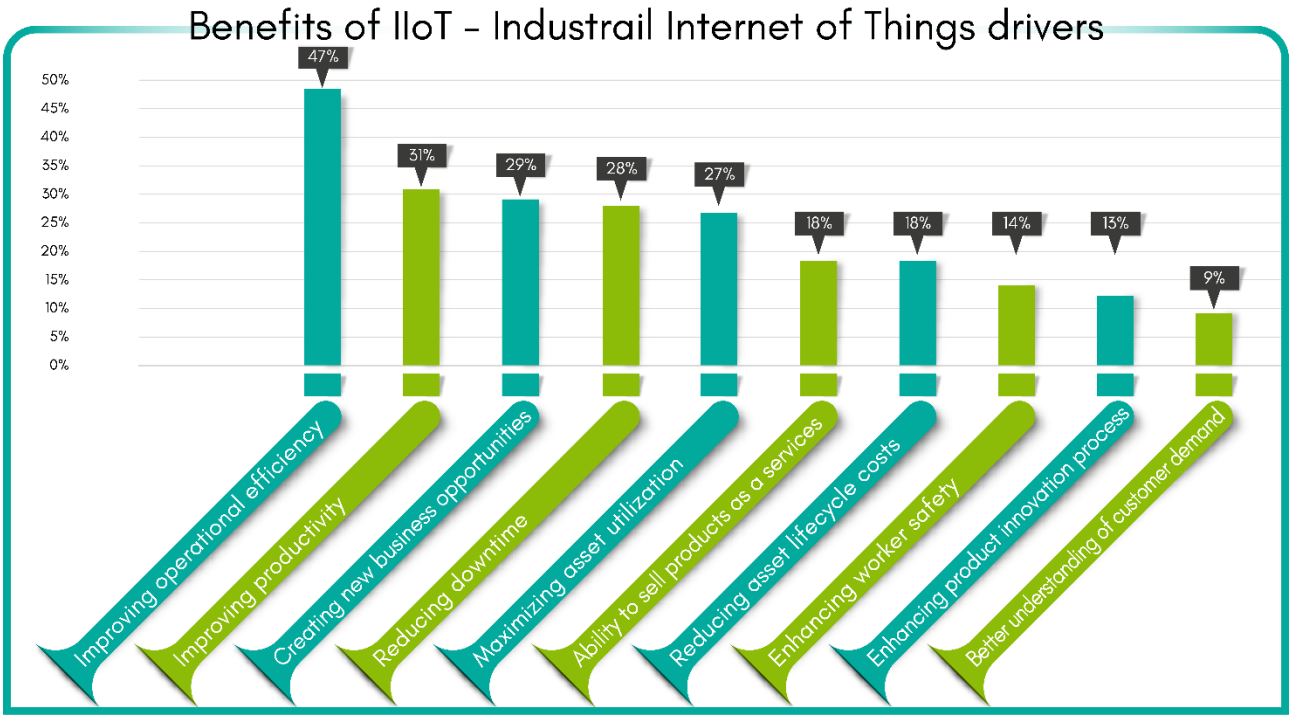

Figure 3: Benefits of IIoT – The Industrial Internet of Things drivers (Morgan Stanley, 2018)

2.3 IIOT ARCHITECTURE

The architecture components described by Pena et al. (2017) include the object-level layer or physical layer, the connection layer, middleware layer and, lastly, the client layer. A depiction hereof is in Figure 4.

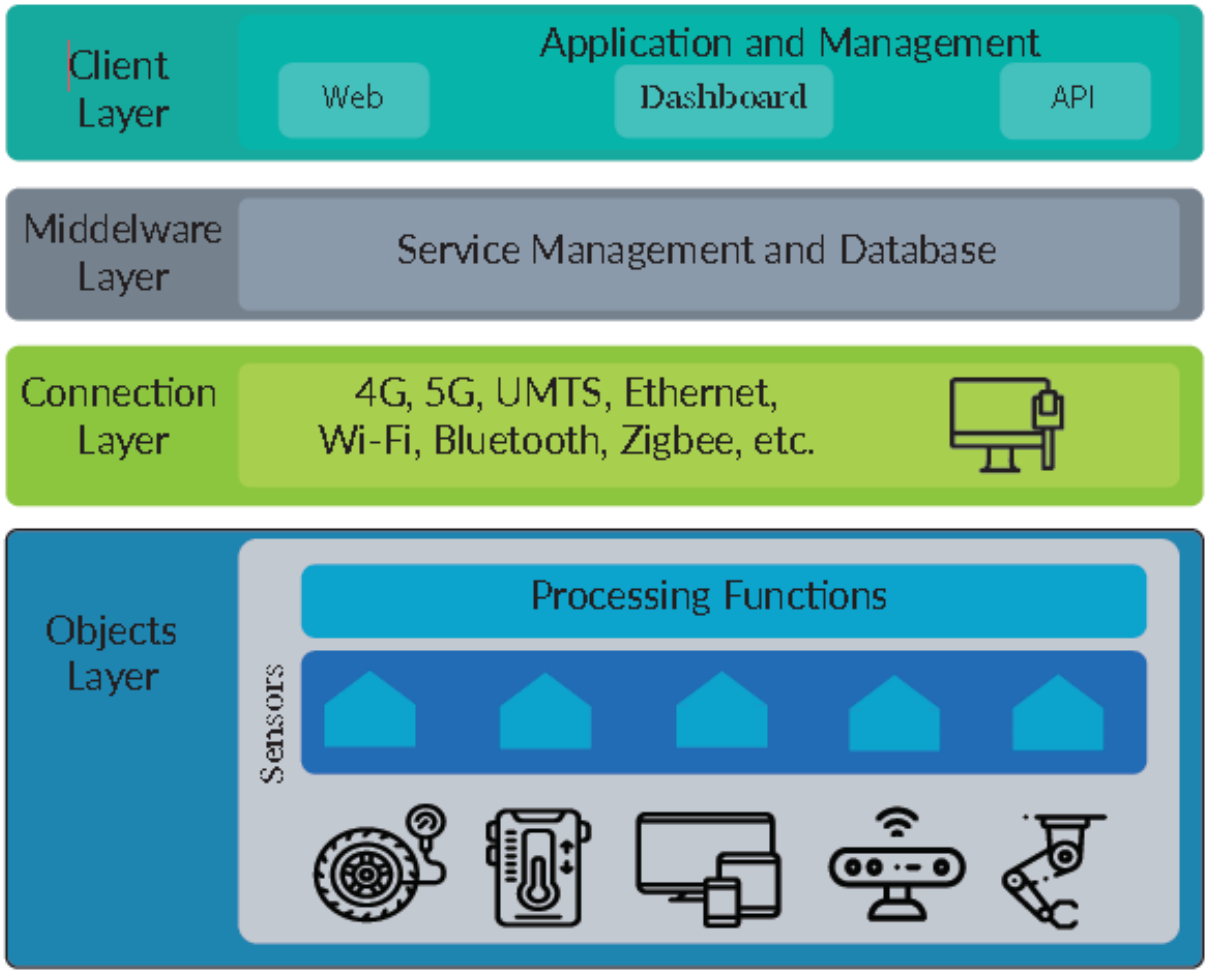

Figure 4: The general architecture of the Internet of Things (Pena et al., 2017)

Boyes, Hallaq, Cunningham and Watson (2018) name the IIoT architectural components as the sensors/actuators (wired or wireless), the internet gateways/data acquisition systems data aggregation, A/D measurement and control, the Edge IT (analytics pre-processing) and, lastly, the data centre/cloud (analytics, management archive). Holdowsky et al. (2019) describe the IIoT as seen in Figure 5. The information value loop comprises a create stage, a communicate stage, an aggregate stage, an analyse and an act stage. In the create portion, the sensors would generate information on events. The communicate stage transfers the data while in the aggregate phase. There is a grouping of information for

analysis purposes. From the new insights gained, the organisation would need to act upon the information.

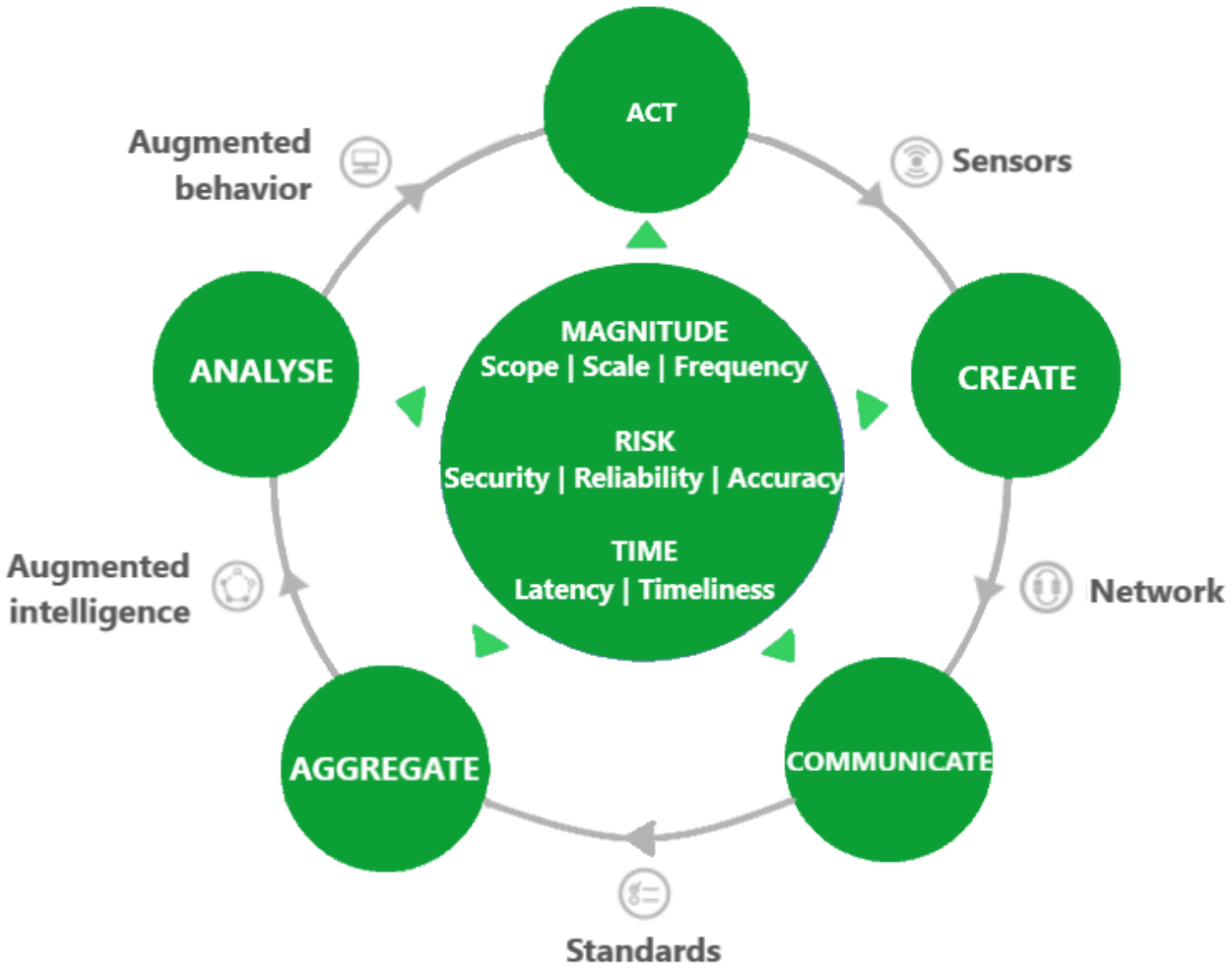

Figure 5: The information value loop (Holdowsky et al., 2019)

The researcher notes that these different architectural components would need to be implemented successfully for a successful IIoT. These components are reviewed further in the following subheadings to extract implementation guidelines from the literature within Chapter 4.

2.3.1 PHYSICAL LAYER OR THE SENSOR LAYER

The “things” or the physical objects are part of the bottom layer. At this layer, the allocation of the identity to the “things” takes place. Also, at this layer, the data are generated with the help of these sensors. These sensors can register and measure changes of elements such as position, motion, velocity, acceleration, force, pressure, flow, acoustic, humidity, light, radiation, temperature, chemical and biosensors (Fraden, 2010). Sensors are used in IoT devices to perceive what is going on around us (Biddlecombe, 2009). Fraden (2010) mentions that the choice of sensors should be according to the application thereof. For instance, when measuring the temperature of something, one would use a temperature sensor. Besides this rather obvious guideline in determining suitable sensors, Fraden (2010) suggests a consideration of the accuracy, repeatability, range, noise, resolution and selectivity of the sensors needed in the implementation of an IIoT solution.

As part of an IIoT implementation, the three factors in consideration to sensors are the price, capacity and the size of the sensors (Raynor & Cotteleer, 2015).

The price in sensors decreased, making it more affordable to deploy sensors at more measuring points. For example, in 2006, an accelerometer price was R28 vs R5.60 in 2015 (Simpson & Lamb, 2014). The capacity of the sensors refers to smarter sensors that function in a more extensive system of power sources, microprocessors and devices (Raynor & Cotteleer, 2015). Smaller sensors enable the implementation of these sensors in more areas that previously did not have the space to do so (Raynor & Cotteleer, 2015). Sensor challenges include power consumption, the security of sensors and interoperability. Sensors are powered through connecting to the electrical grid or batteries (Singh, K. S., Singh, M. P. & Singh, D. K., 2010). In-line power sources are dependable but might not always be practical due to sensor placement. In these cases, batteries would supply the power to these devices. Battery life, efficiency, recharging and replacement might represent challenges in these cases (Holdowsky et al., 2019).

2.3.2 CONNECTION LAYER, THE INTERNET GATEWAYS / DATA ACQUISITION SYSTEMS OR COMMUNICATE LEVEL

The connection layer is the link from the physical objects layer ("things") to the middleware layer (storage and analysis). An essential aspect of the network is to be able to identify each of the things that are connected to the network with a unique address (Holdowsky et al., 2019). Network protocols, be it open or proprietary, are a set of rules that define the way computers identify other computers on a network (Holdowsky et al., 2019). The advantage of open protocols is that it ensures communication across different devices (Holdowsky et al., 2019). Kaur and Monga (2014) classify network technologies as wired and wireless. The decision to use a wired or wireless solution is dependent on the application and proximity of the measured area.

In the selection of network technology, the data transfer rates, cost of internet transit services, adoption of IPv6, interconnectedness, network penetration, security and energy requirements are essential in selection thereof (Holdowsky et al., 2019).

The current 4G and 5G technologies possess a higher data transfer rate, whereby Bluetooth and low-power wi-fi technologies are more applicable to energy-constrained devices (Holdowsky et al., 2019). The 4G and 5G technologies constitute a trade-off between either a high transfer rate or a more energy-efficient solution. It is noteworthy that there is limited penetration of LTE technologies, with 5G still to become available and LTE that accounts for only 5% of all the mobile connections in the world (GSMA Intelligence, 2014).

Internet transfer cost, as an adoption determinant for IIoT, refers to the cost that an internet service provider would ask for transferring data from one point in the network to another point. The IPv6 protocol holds higher scalability capacity for IIoT devices with up to 3.4x10 unique addresses to individually identify add-ons in the IIoT network in contrast to the only 6 billion supported by IPv4 protocol (Commonwealth of Massachusetts Committee on Information Technology, 1998).

2.3.3 MIDDLEWARE OR AGGREGATE LAYER

The middleware layer stores and analyses object data within the database for the corresponding service provided. This data are needed to make decisions for the operation of the “things” (Pena et al., 2017). The middleware layer is also seen as the data aggregation level from the information value loop perspective as described in Holdowsky et al. (2019). The Cambridge Dictionary (2019) defines aggregation as “something formed by adding together several amounts or things together”. In this case, the things would be the aggregation of the data collected by the sensors to increase the value through an increase in the scale, scope and frequency of data that would be made available for analysis (Holdowsky et al., 2019).

To best achieve data aggregation is to work according to the following standards:

- Technology standards (network protocols, communication protocols and data aggregation standards); and
- Regulatory standards with the purpose to regulate the security and privacy aspects of data in terms of the collection, handling, use and sale of data (Holdowsky et al., 2019).

Technology standards are concerned with network protocols (how machines identify each other), communication protocols (how machines communicate with each other in each different language) and data aggregation standards (the standards for processing the data into useful data).

One of the critical concerns of transmitted information within IIoT is security. IIoT security includes data integrity and privacy (Soldatos, 2016; Zolanvari & Jain, 2015). The Federal Trade Commission (2015) provided Fair Information Practice Principals (FIPP) for the guidance of privacy and security recommendations as follows (FIPP could act as a basis for IIot although not currently included).

- Choice and notice: Data collection entities should always give the users the ability to decide what information that they revealed could be shared. This would not necessarily be the case in terms of IIoT due to the M2M communication.
- Purpose specification and use limitation: Data collection entities should clearly state the purpose of why data are collected and the collection thereof be limited to the purpose.
- Data minimisation: Companies could collect data for a specific purpose only and delete the data after the goal has been reached.
- Security and accountability: Data collection entities are responsible for the security of the data and should have the necessary security in place.

There is a need for standards specific to the IoT environment that still need to be fully developed.

2.3.4 AUGMENTED INTELLIGENCE, AUGMENTED BEHAVIOUR AND THE CLIENT LAYER

According to Pena et al. (2017), this layer hosts the application and management services, which host the web, dashboard and Application Programming Interfaces (API). Information and intelligence are embedded in the coding of devices to support people over a network. System characteristics include embedded, context-aware, personalised, adaptive and anticipatory (Madakam et al., 2015).

Holdowsky et al. (2019) define the client layer as described by Pena et al. (2017) in terms of two components: the Augmented Intelligence that drives informed action and the Augmented Behaviour that by using the data can take action. Holdowsky et al. (2019) has an information value loop perspective. New advancements in this regard are the cloud computing services that contain analytical and reporting tools. Machine learning, big data analytics, automation and artificial intelligence help organisations in obtaining value from their data.

2.4 IMPLEMENTATION SUGGESTIONS

Blanchette (2016) refers to the abundant referrals in the literature to the benefits of IIoT implementation in contrast to the few articles in existence on the implementation of the technology. Implementation includes the different decisions that are needed during the process.

Blanchette (2016) suggests the following seven ways to start with an IIoT implementation:

1. Creating a baseline to consider the current data collection methods, processes, visibility (real-time) and analytics to evaluate the capability for the detection of opportunities and issues;
2. Determining if the organisation utilise any machine learning, big data or automation technologies to assist in the capturing, analysing or transmittal of the data;
3. Establishing open communication between different data sources to enable data consolidation for extracting meaningful value;
4. Consolidating different systems;
5. Considering cloud-based applications;
6. Re-looking and documenting the collaboration of functional groups; and
7. Anticipating and planning for any changes in demography by using technology tools.

Lueth (2016) named the following learnings after an IoT implementation that are important considerations. Firstly, the cultural and organisational changes needed are misjudged. He also mentions that IoT projects take over 18 months longer than what had initially been planned. Thirdly, in most of the cases, there are not the necessary skills in-house. Lueth further learned that security is often an afterthought with protocol translation as one of the leading factors. Lueth lastly states that scalability becomes an issue when devices are moving into the thousands because hardware might not be powerful enough or the data models might be too slow.

2.5 STATUS OF IIOT AND MINING

In assessing the status of IIoT in the industrial organisations, especially mining, Inmarsat (2017) indicates that in 2017 only 12% of mining representatives had implemented IIoT solutions to a degree or to a full extent. The World Economic Forum (1971) predicts that the expected impact of digital transformation on the mining industry is in the region of US$428 billion and US$784 billion. Key contributors to the expected monetary impact could be underwritten to the potential realisation of the advantages in terms of higher production and the ability to optimise equipment as described by Merry (2017). IIoT also has the potential to address the challenges of the South African mining industry, as mentioned in section 1, with the research of O'Conner (2017) as:

- A reduction of water and energy resources
- Employee health and safety
- Improved production by the use of automation
- Reduction in waste

Within these industrial operations, there is already an abundance of data that is accumulated with sensors already implemented in existing mining operations – or better known as brownfields sites. The challenges of most of these data are that these data are stored in different systems hindering integration across systems (Mining Review, 2019). This data needs horizontal integration across the value chain, vertically from the field to the control and from planning to maintenance (Mining Review, 2019).

2.6 CONCLUSION

Sensors measure temperature, vibration, motor ampere and speed between themselves with the help of M2M communication. A vast amount of sensor data within the industry feeds data that is connected to a central database. This could be data in the cloud or a local site data pool that is accessible to technology like big data analytics, machine learning, artificial intelligence and automation that could transfer it to usable information for the organisation. The above-mentioned, in short, sums up the IIoT. This IIoT information could assist

organisations in determining inefficiency within processes, improve the personnel worklife by a reduction of effort on lesser essential aspects. This information could also assist in the identification of problems at an earlier state with subsequent business improvement.

Observations that are evident from the literature include:

- Varied definitions of IIoT exist based on the viewpoint of individuals and organisations. Universal standardisation is needed.
- The researcher did not find specific literature on a set of guidelines for an IIoT application that could guide new implementations. There are bits and parts of recommendations that could be used as a basis for the compilation of guidelines.
- Existing academic literature on the topic of IIoT is not readily available due to the rather new nature of it.
- Extracted guidelines would need to contain some of the constructs or architecture of the IIoT and reference to value for the organisation.
  The IIoT would need to contain the physical architecture of the IIoT, as described in Figure 3.

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Chapter 3
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Chapter 4
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Chapter 5
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CHAPTER 3: RESEARCH METHODOLOGY

3.1 INTRODUCTION

Chapter 2 focused on various aspects within the IIoT, technologies, the value of IIoT within the industrial environment and on IIoT implementation recommendations. The purpose was to obtain answers in the existing literature to the research questions that were raised in Chapter 1 by researching the existing literature on the field and the topic of study. From the literature review and the introduction, one could see that there is value in an IIoT implementation within the mining industry. The challenge, however, was obtaining enough relevant sources to compile a set of clear guidelines for the implementation of IIoT in the industrial industry.

Since there is limited literature on IIoT guidelines within the industrial industry, the researcher conducted primary research in the quest to answer the research questions about IIoT implementation guidelines that have been left unanswered, draw a conclusion after data analysis and produce a proposed set of guidelines together with the answers to the research questions.

The following section will describe the research method that produced the necessary data. Further descriptions on the research design, the research instruments, data analysis and the ethical considerations then follow. The research method was used in driving the research to obtain the relevant information to answer the research questions within the following chapter.

3.2 RESEARCH DESIGN

3.2.1 RESEARCH MODEL

The model that was used in the study was based on Saunders,M. Lewis, P. & Thornhill's (2013) five-layered onion (Figure 6) approach that defines the research stages for an effective research methodology. The outer layer or the starting point defines the research philosophy that requires the options in the research approach in Layer 2. Layer 3 entails research strategies, while Layer 4 comprises time horizons, and the most inner layer the data collection methods. Exploring the research onion through the following stages determined the research methodology to conduct the primary research.

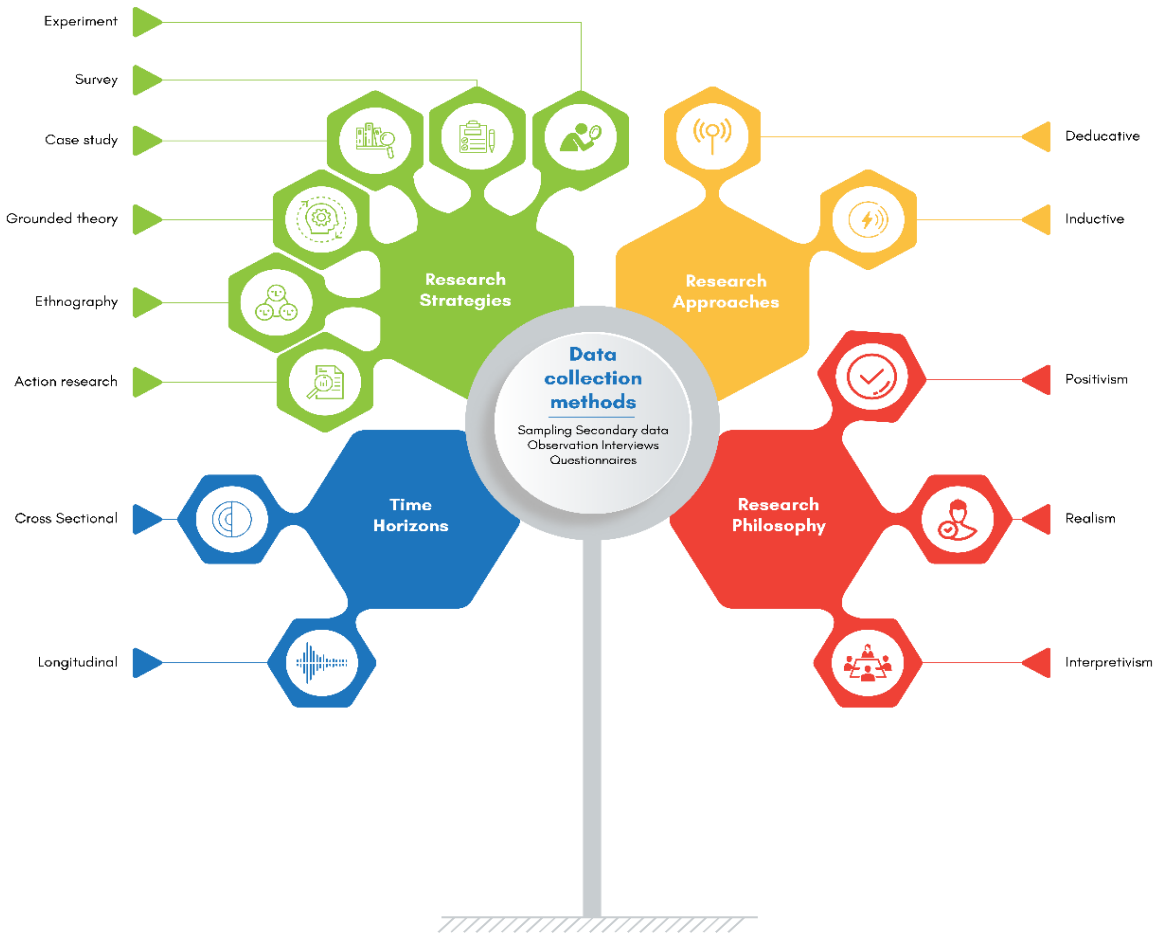

Figure 6: The research onion figure (Saunders et al., 2013)

3.2.2 RESEARCH PHILOSOPHY

The research philosophies used in information systems include positivism and interpretivism. The following overview explains these two research philosophies in better detail.

Positivism

The positivism philosophy believes in a single reality of any situation or phenomenon, decoupled from any researcher's belief (Hudson & Ozanne, 1988). For this reason, any research conducted would be in a structured and controlled fashion with a precise research topic and a suitable research methodology (Carson, D., Gilmore, A., Perry , C. & Gronhaug, K., 2001). Furthermore, there would be a clear distance between the researcher and the researched to avoid any findings that could be affected by emotion, reason and feeling (Carson et al., 2001). They would use clear, consistent logic with mathematical and strategical techniques to uncover the reality (Carson et al., 2001).

Interpretivism

Researchers who want to understand situations according to the significance attached to them by society would conduct interpretive studies (Myers, 2012). Acquired knowledge within the interpretivism field is "socially constructed rather than objectively determined" (Carson et al., 2001). The research adopted a more personal flexible research structure (Carson et al., 2001). The research was conducted with a prior insight of the context but with the understanding that it needs further investigation because of the complexity and unpredictable nature of one's understanding of reality (Hudson & Ozanne, 1988).

When the researcher considered the research philosophy that would be best suited for the research, one of the determining factors was the practical implication of the research. Due to the new advances in terms of sensor availability and big data analytic tools, the value of IIoT implementations in the mining arena became feasible only during the last few years. This means that there are not many cases of IIoT implementation. This study relied on

qualitative data and was interpretive of nature; therefore, the research philosophy was interpretivism.

3.2.3 RESEARCH APPROACH

Further to the previous discussion, consideration was also given to the different research approaches. A deductive approach would need a pre-existing theory to develop a hypothesis and to formalise a research approach to test it (Siverman, 2013). An inductive approach is described as a “move from the specific to the general” (Bryman & Bell, 2011). In this study, an inductive approach was followed where the guidelines were derived.

The quantitative research approach entails large-scale survey research to generate statistics. It makes use of questionnaires or structured interviews, whereas the qualitative research approach would explore experiences through interviews or focus groups. It focuses on an in-depth opinion from the participants (Dawson, 2009). In this study, a qualitative research approach was followed.

3.2.4 RESEARCH STRATEGY

According to Saunders et al. (2007), the research strategy refers to how the work is carried out by the researcher. Within the research philosophy, it was clear that the philosophy would be interpretivism and the research approach qualitative. With this in mind, the research strategy should be in line with the interpretivism philosophy. The following analysis was done on the different strategies to acquire the most suitable for interpretivism.

Experimental studies as research strategy relate to the study of causality. During experimental studies, a group of subjects are exposed to a cause. During the same experimental studies, a second group is not exposed to the same cause. An observation of the effects between the two is done to consider the differences (Bhattacherjee, 2012).

Experimental studies are more aligned with the positivist paradigm and are therefore inappropriate.

Field surveys, as a research strategy, focus on a sample of subjects in a field set-up using a questionnaire or structured interviews for capturing the conditions at a point in time (Bhattacherjee, 2012). Field surveys require questionnaires and structured questions (Saks & Allsop, 2012). Field surveys are not ideal for research on IIoT either.

Secondary data analysis examines the data that were collected from other sources (Bhattacherjee, 2012). This was used earlier on in the literature review section of the dissertation and helped with the answering of some of the research questions, but secondary data analysis is a positivist strategy and therefore not in line with the chosen interpretivism philosophy.

Action research, as a data analysis tool, is a positivist strategy and is deemed not applicable. Action research is an introduction of “actions”/”interventions” in processes to test if an anticipated outcome will be correct. These actions are based on the theory that explains the interaction between change and action (Bhattacherjee, 2012).

Ethnography, as a data analysis method, requires lengthy timelines. Ethnography would not be possible as a research strategy because the researcher is introduced to the society in which he needs to do research. The period for the researcher to study the culture of the social portion is between eight months and two years (Bhattacherjee, 2012).

Case studies, as a research strategy, can be quantitative or qualitative. On the other hand, positivist and interpretive (Yin, 1984) case studies are conducted in real-life environments over some time. Data collection takes place through interviews, questionnaires and reviewing of pertinent documentation (Bhattacherjee, 2012).Yin (1984) further explains a case study as a data collection technique used for the investigation “in context or in its real-world setting where cases are selected based on the research purpose” (Yin, 1984). A case study will be used on previous successful IIoT implementation cases within an industrial

environment to obtain the necessary data and to investigate the research questions. Case studies are in line with the interpretivism philosophy.

According to Hofstee (2015), in using case studies, the focus is on a single case, for example the successful implementation of an IIoT for the deduction of principles that could be of value in similar future cases. Hofstee further mentions that case studies are useful when there is a need for in-depth knowledge for whatever reasons. Beiske (2002) mentions that case studies are usually qualitative research methods to generate possible theories by observation of reality. Adelman and Jenkins (1977) regard a case study as “an umbrella term for a family of research methods having in common the decision to focus an inquiry around an instance”. A case study means that the focus of the researcher would be on a small number of cases or a single case (Beiske, 2002).

Strengths

Gill and Johnson (2002) describe the strengths of a case study as a valuable tool if little is known on a topic and it would be difficult for a researcher to find data on that topic. In that case, the study of a case and the analysis of it would enable the researcher to gain new knowledge on the topic. Additionally, Gilbert (2001) mentions that a case study allows studying a specific case or situation intensely, and it can relate the findings to the study. Case studies would be useful in testing the findings. The view of Nisbet and Watt (1980) is that, in some instances, only a real situation would be able to portray the full picture of a study. Nisbet’s view is very relevant in the case of an IIoT implementation. Only if a person has previously implemented an IIoT solution in a mining organisation would the study thereof be able to shed light on an IIoT implementation and could the study findings be tested in a future IIoT implementation in a similar environment. Another positive aspect of case studies is that reviewing differences and similarities between cases in essence (in this instance implementations of IIoT in mining) increases the validity of the theory of the study (Eisenhardt, 1989).

\

Limitations

The limitation as described by Eisenhardt (1989) is that building the theory from a case does not often lead to “grand theories” but it instead “stays modest”. He adds that one can use case studies for thorough testing of the theory.

Benbasat et al. (1987), in his review of case studies, summarised the following weaknesses:

- A lack in terms of the detail of how data were collected;
- Lack of clear research purpose and objectives;
- Not enough use of triangulation in increasing the reliability of research;
- Limited use of longitudinal methodologies; and
- Not enough details on the data sources used.

3.2.5 TIME HORIZON

The two time horizons, namely cross-sectional and longitudinal timeframes within the fourth layer of Saunders et al.’s (2007) research onion, refer to the needed time for completion of the research. Within the cross-sectional timeframe, the collection of data would occur at a certain point (Flick, 2011). Data collection in this manner would be the study of a particular case at a specific time. On the opposite side, Goddard and Melville (2004) describe longitudinal time as the repeated collection of data over a more extended period, thus monitoring change over time. The time horizon was a cross-sectional timeframe.

3.3 DATA SAMPLING

3.3.1 SAMPLE SIZE

Sample size refers to the number of selected respondents from the population (Newman, 1998) in the case study. The sample size focuses on an IIoT implementation within the South African Industrial industry. Approached respondents were the companies that were involved in the implementation of the IIoT in the industrial area.

3.3.2 SAMPLING TECHNIQUES

Sampling techniques refer to the selection of the sample size in the overall population (Newman, 1998). Different techniques include random sampling, where participants are chosen randomly within a society. Random sampling was not applicable in the researcher's case due to the specific nature of the study; a stratified sample would ensure that the proper representation from the population is chosen and would, therefore, apply to the study (Newman, 1998). Convenience sampling would be from an existing framework, for instance and educational institute, for when the researcher might need the opinion of students.

The two main sampling techniques are probability and non-probability sampling (Lewis et al., 2000). A discussion follows:

A. Probability or representative sampling is mostly a survey-based technique for the selection of random responders.
B. Non-probability or judgmental sampling offers the researcher the prospect of selecting a purpose-driven sample and reaches less identifiable members within the population. The technique also offers alternatives within the non-probability sampling technique at the discretional judgment of the researcher, including:

   i. Purposeful sampling is when only the judgment of the researcher selects the sample.
   ii. Snowball sampling technique would apply when it is challenging to identify respondents.
   iii. Convenience sampling occurs when the researcher opts for cases that are the easiest to obtain the data sample.
   iv. Self-selecting sampling takes place where an individual indicates their desire to partake in research.
   v. Quota sampling or statistical analysis is enabled if there are sufficient responses from a quota or a more substantial population.

Purposive sampling was the technique chosen for this study. Purposeful sampling is a qualitative research technique that finds information-rich cases and simultaneously offers the effective use of limited resources (Patton, 2002). Purposeful sampling means that the researcher would recognise and hand-pick people with specific knowledge and experience in the field under study (Creswell & Plano Clark, 2011). The importance of participants' willingness to participate and avail themselves also adds to the benefits (Bernard, 2002). In this case, it would be individuals who have successfully implemented an IIoT within the industrial organisation within South Africa. The respondents consisted of IIoT architects, IIoT platform suppliers, IIoT communication experts, implementations companies, IIoT project managers and IIoT technicians.

3.4 DATA COLLECTION METHODS

3.4.1 RESEARCH INSTRUMENT: LITERATURE REVIEW

Literature reviews as a research instrument enabled the researcher to build the research on previous findings from other researchers on the topic of the IoT and IIoT. This entailed the meticulous consideration of relevant literature to gain a clear understanding of the research topic. The literature review further aided the researcher in answering the research question. Baumeister and Leary (1997) describe a literature review as a systematic method of synthesising and collecting previous research. Webster and Watson (2002) wrote that a well-conducted research review creates a firm foundation for the enablement of theory development.

3.4.2 RESEARCH INSTRUMENT: INTERVIEW

The source of the primary data would be data extracted from respondents during interviews (Bryman & Bell, 2011). The research instrument in use was interviewing for the collection of primary data. This technique is most often employed when conducting qualitative research (Bryman & Bell, 2011).

Lofland and Lofland (1994) and Kvale (1996) state that interviews in its most basic form will be a "guided conversation". Lofland and Lofland (1994) and Moser and Kalton (1971) describe interviews as "conversations between interviewer and interviewee, but with the specific aim to gain valuable information from the respondent".

The analogy used by Cohen (1976) and Bell (1999) to describe interviews is that of a fishing expedition as it requires "careful preparation, much patience, and considerable practice if the eventual reward is to be a worthwhile catch". The researcher interviews to establish a new theory in a specific research area and to gain a data-driven (inductive research) start, analysis of the data and a conclusion on prior theory (theory-driven) (Perry, 1998).

Strengths

Interviews offer adaptability compared to questionnaires with their fixed information queries in the sense that interviews offer the opportunity to probe responses, clarify issues and delve deeper into responses (Bell, 1999). They can offer valuable in-depth knowledge on how situations are perceived, which would not be possible without little or with no interaction. The interviewer can simultaneously also assess the interviewee's underlying attitude (Gilbert, 2001).

Limitations

Cicourel (1964) mentions the problems that could arise if there are trust issues between the interviewer and interviewee, and the influence that it could have on meaningful answers. The other side of the coin would be the ability to scale the research to more respondents, which would entail an exponential increase in cost to obtain the primary data (Perry, 1998). Prejudice on either side of the interviewer or interviewee could exist (Saunders et al., 2007), concluding that it is a subjective data collection method influenced by the relationship between the interviewer and interviewee with influences from cultural differences. Bell (1999) envisions a problem in terms of the analysis of the findings of the interview due to the inability to quantify the analysis since it would produce qualitative data.

Three main interview types are open-ended or unstructured interviews, focused or semi-structured interviews and structured interviews (Myers, 2012). Edwards & Holland (2013) state that the use of structured interviews occurs in quantitative research to ensure all respondents are asked the same question in the same order. An open-ended interview is a flexible conversation to recover facts and gain an understanding of personal opinions. An open-ended interview enables the respondent to articulate his full meaning around the subject and to let the respondent think freely about the topic. The hybrid middle way that has some of the best of the interviews is a semi-structured interview. A semi-structured interview uses a framework of investigation. Themes are used for questioning respondents with flexibility, keeping in mind the direction of the conversation (Edwards & Holland, 2013).

The University of Adelaide (Permani, 2018) uses the following farm example in Figure 7 to simplify these concepts:

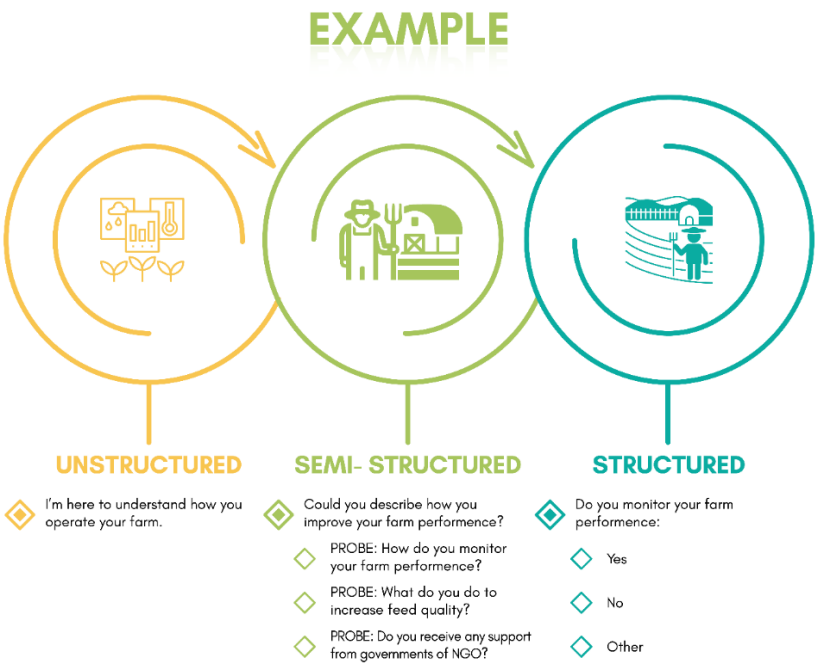

Figure 7: Examples of differences between unstructured, semi-structured and structured interviews

Source: Permani (2018)

Employing the semi-structured interview as a research instrument, the researcher expected to gain the necessary data to answer the research questions. The added benefit was that an IIoT implementation platform could emerge from analysing the data. The researcher had a themed list with relevant questions designed to gather data on the specific field (Welman & Kruger, 2001). In this study, the researcher believed that interviews were suitable because there is a need for people's expert opinion and knowledge in a specific field. The researcher was interested in the facts and description of the process. Interviews allowed the respondents to express themselves with the additional provision of more information.

Open-ended questions were asked (Table 2) in the hope that participants would elaborate on their responses, while the researcher recorded their answers.

Table 2: Question bank directed at extracting implementation guidelines and critical considerations for the IIoT

Interview question category	*Sub-category*	*Question number*	*Interview question*	*Purpose of question*	*How will this help to answer the research question?*	*Expected result*
Understand the interviewee		1A	What is your primary job title?	To establish credibility that the interviewee has the experience in the IIoT field.	Experienced IIoT implementation resources would provide expert opinion and trustworthy answers.	The interviewee is working on IIoT implementations.
		1B	How long have you held this position?	To ascertain the working hours that the interviewee spent in these types of implementations and the amount of experience in the field.	Experienced IIoT implementation resources are needed for expert opinion and trustworthy answers.	The higher amount of industry-specific practical experience would be more credible.
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	To ensure that the understanding of the interviewee is in line with the definition of IIoT. The purpose is to establish credibility with the interviewee's answers and to establish him as an expert in the field.	Need experienced IIoT implementation resources for extracting clear IIoT guidelines.	The interviewee has a basic understanding of what IIoT is.
		2B	What do you see as industrial organisations?	To establish that the interviewee worked within industrial organisations.	Need for industry-specific use cases for the compilation of industry-specific IIoT guidelines.	The interviewee should have a basic understanding of what industrial organisations entail.

		2C	Did you implement any IIoT solutions for industrial companies?	To establish that the interviewee implemented IIoT in the industry that this dissertation is focusing on.	Need for industry-specific use cases for the compilation of industry-specific IIoT guidelines.	It is essential that the interviewee implemented IIoT solutions.
	Existing guidelines	3A	Are there any formal guidelines used in implementing the IIoT in the industrial organisation?	To establish formalised guidelines that were not evident from the secondary literature review research.	The research is based on guidelines for IIoT implementations.	The expectation is that there are no formal guidelines in place.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	To establish if the formal guidelines are adequate.	The research is based on guidelines for IIoT implementations.	The expectation is that there are no formal guidelines in place.
	Offered guidelines	3C	Are there any informal guidelines that are used in implementing the IIoT in the industrial organisation?	To help the establishment of guidelines for the IIoT implementations.	Suggestions of guidelines for an IIoT implementation would become available.	The interviewee offers guidelines.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	To help the establishment of guidelines for the IIoT implementations.	Suggestions of guidelines for an IIoT implementation would become available.	The interviewee offers guidelines.
Key considerations before the implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	Defines the landscape that would benefit from IIoT implementations. It might be possible that specific industries do not need any such implementations.	This highlights the key considerations.	The interviewee offers industry insights.

		4B	In your opinion, what needs to be in place for an IIoT implementation?	Defines a necessary architecture that is in place.	This highlights the key considerations.	The interviewee offers the architecture that needs to be considered.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	Recognises existing architecture.	This highlights the key considerations.	The interviewee considered existing architecture and the role that it plays in an IIoT solution.
	Insights about the Architecture of IIoT	5A	Please draw the architecture needed for IIoT implementation.	Presents the key considerations and components for an IIoT implementation visually.	This highlights the key considerations.	Hope to have a view of the architecture for implementation guidelines.
		5B	In your opinion, what are some of the critical considerations for any industrial organisation embarking on an IIoT implementation?	Defines other aspects that are not evident from the architecture.	This highlights the key considerations.	Unknown what to expect.
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done	The case needs to provide context.	This aids with the lessons learned for new IIoT implementations and would be	Insight on lessons learned.

			differently given the same situation.		considered in the IIoT implementation guidelines.	
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	The interviewee offers considerations for IIoT implementation guidelines.	This aids with the lessons learned for new IIoT implementations and would be considered in the IIoT implementation guidelines.	Insight on lessons learned.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	The interviewee offers considerations for IIoT implementation guidelines.	This aids with the lessons learned for new IIoT implementations and would be considered in the IIoT implementation guidelines.	Unknown what to expect.
		6D	What would you describe as the advantages of an IIoT implementation?	The interviewee offers considerations for IIoT implementation guidelines.	This aids with the lessons learned for new IIoT implementations and would be considered in the IIoT implementation guidelines.	Unknown what to expect.

3.5 DATA ANALYSIS

Within a qualitative study, analysing data would entail the synthesis of information that was obtained by interviews with a coherent description of what was observed (Guetterman, 2015). De Hoyos and Barnes (2018) note the following six-point general analysis process:

- Add initial codes
- Add comments/reflections
- Look for patterns, themes, relationships, sequences, differences
- Explore patterns
- Elaborate and make small generalisations
- Link generalisations to the body of knowledge to construct.

Following these guidelines, the researcher planned the following actions to analyse the data. Interviewee data were captured via a voice recorder and transcribed verbatim via transcription software to not lose any of the respondents' ideas, knowledge or intended data. The allocation of codes to data and similar data that were present in other transcripts received the same code – as part of the analysis, writing of comments and notes. The sorting of data occurred according to themes. The data were then grouped by theme according to the "thematising" idea of Mitchell & Jones (2004). The themes were then further explored, elaborated upon and linked to the body of knowledge construct.

Reliability and validity

Validity within the research refers to the ability of the instrument to demonstrate that it is finding out what it should. Reliability means that the analysis procedure consistently yields the same finding (Teherani, A. Martimianakis, T, Stenfors-Hayes, T. Wadhwa A, & Varpio, L., 2015).

3.6 ETHICS

Ethics should be considered during all facets of research as ethical issues could arise at any given moment within the research (Bryman & Bell, 2011). If ethical behaviour in the research is disregarded or neglected, it would directly impact the integrity of the research. Brannick, Levine and Morgeson (2007) summed it up with the following words: “Not doing harm, not breaching confidentiality, not distorting the data.” The summary of Brannick et al. is in line with Bryman and Bell’s (2011) principles of informed consent, avoiding harm to the participants, invasion of privacy and deception.

The researcher committed to ethical research by adhering to the following guiding principles:

Principle 1: Informed consent means that “prospective research participants should be given as much information as might be needed to make an informed decision about whether they wish to participate in the study”.

Principle 2: Avoiding harm to the participants. Bryman and Bell (2011) state that society mostly regard it as unacceptable if one were to cause harm to another. Bryman and Bell define harm as any “psychological or physical disturbance to participants” and mention specific cases of harm as problems to self-esteem, stress, harm to career or causing the responders to do illegal or unethical acts.

Principle 3: Invasion of privacy. Bryman and Bell (2011) give the clear guideline that one would comply with ethical behaviour if one respects the participants’ privacy by valuing the participants’ choice to answer a specific question or not and that they would not feel one is intruding their privacy.

Principle 4: The principle of deception. Bryman and Bell (2011) explain deception to be when a researcher represents their work as something that it is not. The researcher notes that

these certain principles are of the utmost importance and should be the guiding factors when conducting the research.

Further to this when conducting ethical research, it should be measured. Edwards & Holland (2013) suggest the following three points for measurement within qualitative research: trustworthiness, quality and rigour.

For trustworthiness and data quality, the collection of data was from experts in the relevant field. These experts were directly involved in the research topic. It is noteworthy that the semi-structured interview followed a rigid structure of interview themes and data would be immediately transcribed. The achievement of data quality was by pre-testing interview questions before conducting the actual interview.

3.7 CONCLUSION

Chapter 3 revised the different research methods to find an approach to conduct the study. This chapter afforded the researcher the ability to select the most applicable and appropriate research method to the field of study. A summarisation of the selected method is tabled in Table 3.

Table 3: Research approach summary

RESEARCH APPROACH SUMMARY	
RESEARCH PHILOSOPHY	Interpretivism
RESEARCH APPROACH	Deductive
RESEARCH DESIGN	Qualitative
RESEARCH STRATEGY	Case study
TIME HORIZON	Cross-sectional
DATA COLLECTION TECHNIQUES	Literature review, interview

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Literature Review
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CHAPTER 4: RESEARCH FINDINGS

4.1 INTRODUCTION

The preceding chapters provided the different definitions of the IIoT and its applicability for future use and value in the industrial environment. It reviewed two deep architectures of the IIoT together with the advantages and disadvantages. The literature review provided a theoretical overview of the concept of IIoT. Chapter 3 presented the different research methods, and an applicable appropriate approach was identified to conduct the study. The basis for the research is the qualitative study research method that would be achieved by literature studies and interviews.

In Chapter 4, an analysis of the research findings of the research questions will be presented to derive new learnings that could contribute to new knowledge in the academic field. The basis for analysis was the collected interviewee data and the findings within the literature. The problem statement, research questions and purpose are repeated below for the reader's benefit.

4.1.1 PROBLEM STATEMENT

The problem due to the IIoT sensor hardware that only recently became affordable, advancements in the powering of these sensors and the development of the analytic tools on an application layer was that there are only a few completed IIoT implementations in the industrial environment. Because of the few IIoT implementations, there is a gap in the availability of literature or guidelines that could assist companies in implementing an industrial IIoT solution. Without proper guidance or assistance from previous lessons learned, implementers need to discover how to implement IIoT solutions in the industrial environment. If there are no proper guidelines defined, it could lead to overruns in project, cost, time and scope. In addressing this problem, the research focuses on delivering a set of guidelines that could be to the benefit of industrial organisations for the implementation of IIoT solutions. The research questions used in this research are described next.

4.1.2 THE MAIN RESEARCH QUESTION

What is a list of guidelines that can be used for IIoT implementations?

4.1.3 THE RESEARCH SUB-QUESTIONS

What guidelines are pertinent in the existing literature?

What are the critical considerations for any organisation planning to implement the IIoT to gain the advantages?

What are the lessons learned from other implementations in industrial organisations that could assist new implementations?

4.1.4 PURPOSE

The purpose of this research was to assist industrial companies that want to create further value in their operations with the help of the IIoT with a set of guidelines that could inform the organisation. Industries could find it challenging to navigate through IIoT information to

understand what needs to be in place for an IIoT implementation and the building blocks of an IIoT solution. If there are available implementation guidelines during the planning phase of an IIoT solution for an industrial organisation, the organisation could use these guidelines as information on what an IIoT solution entails. Guidelines could assist the prospective IIoT implementers in creating a complete scope based on prior implementations. A clear scope could mean that they could do better costing of an IIoT solution. The organisation could establish timelines and prevent unnecessary rework. Guidelines could further inform the organisation if they would indeed want to pursue an IIoT implementation.

4.2 DATA INTERPRETATION AND ANALYSIS

The obtaining of data occurred in two ways. A literature review was conducted to establish available guidelines. There were recommendations published in the literature that could assist as data input. Interviews with experienced professional implementers of IIoT were also conducted. The interviewees based their knowledge on cases that they implemented. The data gathered will be discussed next.

4.2.1 LITERATURE REVIEW

In the quest to extract guidelines, the literature survey answered the **research sub-question,** namely: What guidelines are pertinent in the existing literature? The inclusion of these literature recommendations, as described in 2.4 in the literature review, would be needed to complete a set of guidelines. These recommendations, as suggested by Blanchette (2016), are as follows:

- Create a baseline to consider the current data collection methods, processes, visibility (real-time) and analytics to evaluate the capability for the detection of opportunities and issues;

- Determine if the organisation utilises any machine learning, big data or automation technologies to assist in the capturing, analysing or transmittal of the data;
- Establish open communication between different data sources to enable data consolidation for extracting meaningful value;
- Consolidate different systems;
- Consider cloud-based applications;
- Re-look and document the collaboration of functional groups; and
- Anticipate and plan for any changes in demography by using technology tools.

Lueth's (2016) recommendations could be seen as complementary to the suggestions of Blanchette (2016) and also need to be considered when compiling implementation guidelines. Lueth's recommendations are that the cultural and organisational changes needed are misjudged. These changes need careful management.

- IoT projects take at least 18 months longer than what had been initially planned. Better planning is needed.
- In most cases, there are not the necessary skills in-house.
- Security is often an afterthought with protocol translation as one of the leading factors.
- Scalability becomes an issue when devices are moving into the thousands because hardware might not be powerful enough, or the data models might be too slow.

It is also evident from the constructs of the IIoT that without the proper architecture, it would not be possible to implement an IIoT solution. This architecture is essential and should be included as required before any implementation could be attempted. These include all the components listed in Figure 3. The architecture components described by Pena et al. (2017) are:

- The object-level layer or physical layer
- The connection layer

- The middleware layer
- The client layer.

Lastly, Chan (2017) clearly defines ten critical considerations before any IIoT implementation can take place (Table 1), namely:

- A higher level of security because of the impact that security violations could have on manufacturing processes.
- The ability of the IIoT to function with legacy systems with different protocols and data sets.
- IIoT should be able to scale to support thousands of robots, controllers and inputs and purposely build applications. IIoT should take the scheduling, workflow integration, data collection and analysis into consideration.
- A very high level of precision and also a very high level of accuracy.
- The ability to frequently reprogram the industrial systems to support new systems.
- Low latency because of high-speed production systems.
- Reliability of industrial systems and sensors.
- Resilience to failure.
- Any IIoT devices handling automation would need to be able to handle intelligence at the device level.
- Industrial equipment should withstand harsh conditions for years and should be easily maintainable.

Therefore, the implementation guidelines from the existing literature are presented in findings 1–16 presented as follows:

FINDING 1: Before any implementation commences, a clear baseline should be created that considers the current data collection methods and the existing infrastructure vs the needed IIoT architecture.

FINDING 2: If there are cloud services used, the presence of data analytics, the relevant needed reports for decisions and the level of automation should be considered.

FINDING 3: Consider the business demographics.

FINDING 4: Thorough planning is needed before any IIoT implementation can commence.

FINDING 5: Before implementation, consolidation of different source data is necessary.

FINDING 6: Management of cultural and organisational changes is vital.

FINDING 7: Lack of in-house skills for IIoT implementations and the maintenance thereof after.

FINDING 8: Scalability of IIoT devices when they grow into the thousands.

FIDNING 9: Serviceability of the industrial equipment. Equipment should be easily maintainable.

FINDING 10: The level of intelligence needed at the measuring device level and the ability to handle automation on the lower level.

FINDING 11: High resilience to failure of hardware against environmental factors.

FINDING 12: A high level of security to prevent impacts on production and safety.

FINDING 13: Compatibility of newer equipment and systems within IIoT to function with existing or legacy systems.

FINDING 14: IIoT solution should have a very high level of precision and accuracy.

FINDING 15: The industrial systems should be easily reprogrammable to support changing processes.

FINDING 16: IIoT should have a low latency because of the high-speed production systems in use.

4.2.2 INTERVIEWS

The selection and involvement of participants in the interviews were because of their direct connection in the IIoT and the implementation thereof. The specific candidates were

selected based on their expertise on a particular aspect of the IIoT and their role in implementations of IIoT, for instance IIoT architecture specialist.

Each interview followed the same structure as explained with the following steps:

1) The researcher compiled guiding questions and explained the purpose of the study (Annexure A).
2) As a basis, the University of Pretoria approved the ethical consent (Annexure B).
3) The researcher identified leading industry experts in the field of IIoT implementations.
4) The researcher contacted the interviewees telephonically and requested the help of the interviewees.
5) The researcher informed the interviewee and the interviewee's company of the purpose of the study and supplied the interviewee with the questions and schedule (Annexure C).
6) After obtaining approval of the employing company and the interviewee, a decision on a suitable time, duration and location followed.
7) The commencement of the interview after a summary description on the purpose of the research was given. Confidentiality issues were discussed and explained before the interviews proceeded. Ethical consent was provided and signed as per Annexure D. The recording of interviews took place for completeness and transcription purposes.
8) The researcher expressed his appreciation for the interviewee's time and transcribed the audio after each interview.

The interview questions consisted of three different categories, as follows:

Category 1 and a portion of Category 2 consisted to establish the interviewee's experience and credibility in the different fields of the IIoT. The remainder of Category 2 and Category 3 aimed explicitly to extract guidelines for the implementation of the IIoT from the interviewees with the help of assessing the key considerations and the lessons learned within the different categories. The different questions within the interview guide were designed to answer the main research question and the research sub-questions that support the primary research.

Caudle (2004) suggests that sub-questions could be created for each research question to aid in obtaining a clear answer to the research objectives. The duration of the interviews was between 30 and 60 minutes over two months due to the decentralised location of the different interviewees.

The research goal is to communicate the main findings of the study concerning the research questions. Coding and grouping as qualitative data analysis techniques to extract the guidelines were used. This analysis technique is in line with the suggestions of Kumar (2005) that content analysis is an "analysis of the content of an interview identifying the main themes that emerge from responses provided by participants". The use of both interpretive and descriptive coding, as suggested by Caudle (2004), and the categorisation and thematisation of data are attempts to recognise developed patterns and explanations. Developing a predefined set of codes as suggested by Palic, M. Vignali, C. Hallier, B. Stanton, J. & Radder, L (2016) acted as a starting point.

4.3 CATEGORY 1 ANALYSIS OF RESPONDENTS OF THE INTERVIEWS

A. Participants in the study

A total of twelve interviewees were invited to participate. Two felt that they did not have sufficient knowledge due to the relatively new concept of the IIoT. Due to the new nature of IIoT implementations in the South African industries, interviewees that successfully implemented IIoT solutions were challenging to identify.

The researcher noted that the data mostly correlated between some interviewees and, in essence, became saturated from the responses received in the interviews towards the end. Consultation with interviewees from the different depths of the field took place. These interviewees represented the various layers within the IIoT architecture. For instance, the network specialist on IIoT had more precious insight into his area of expertise. All interviews were conducted on a face-to-face basis and recorded.

As noted in the previous chapter (Table 2), the interviewees were selected based on the different aspects of the IIoT. Interviews with IIoT implementation companies were conducted to answer specific questions about IIoT implementation. This recipe was also applicable to communication. Related questions with the help of a network-related IIoT company in South Africa were requested. The same applied to an IIoT platform and the integration of IIoT solutions. The interviews were based on the use of the purposeful sampling approach as described by Bryman and Bell (2011), which is driven by the research questions.

B. IIoT experience analysis

The interviewed respondents' experience ranged from two years to ten years. Interviewees' experience in the field also ranged from implementation business owners to executives. Table 4 contains the demographic analysis of the interviewees as directed by the interviewee guide.

Question 1 A-B, concerning the interviewee's first job title and the duration, provided the data validation of experience and also the credibility of the participants. The purpose of Question 2A-C was to establish that the respondents understood the constructs of the IIoT in terms of the industry based on previous implementation cases. The responses gained in these questions were compared to the known constructs of the IIoT, as found in the literature to ensure that the interviewees understood the constructs of the IIoT.

Table 4: Analysis of interviewees IIoT experience

Participants	Number of years in industrial operations technology	Current designation	Identification of the industrial organisation	Implementation of IIoT solutions
Respondent 1	10 years	Head Technical Solutions IIoT Platforms	Manufacturing facilities, power generation, food and beverage, water, and wastewater. Organisations such as machine suppliers that would use IIoT to monitor their machines remotely.	Implemented solutions and platform software supplier.
Respondent 2	2 years as executive	Communications and Network Provider	Industrial organisations are organisations that produce something.	I assist our channel partners with IIoT solutions for implementation.
Respondent 3	13 years	IIoT System and Infrastructure Architect	Involves manufacturing or converting raw material into something that is an end product.	No, but I design for implementation teams and advise them.
Respondent 4	17 months	Co-founder and Director of an IIoT implementations company	Manufacturing facilities, power generation, food and beverage, water and wastewater. Organisations such as machine suppliers that would use IIoT to monitor their machines remotely.	Implemented solutions
Respondent 5	28 years	Senior Project Manager in the field of Information Management	Every organisation that's part of manufacturing or mining. All the big industries, food, pharmaceuticals.	I am busy implementing several implementations for the organisation that I am employed in at the moment.
Respondent 6	6 years of experience in the field	Solution Business Analyst in Automation	Scalability and size of the organisation, e.g. mining.	Radio communications on decentralised mobile substations that collect data of these substations.

Respondent 7	20 years of experience	Electrical Engineering Technologist in the field of Automation	Companies	Hand-held communication devices, energy consumption information and statistics needed in a central location.
Respondent 8	20 years	Information Management Project Manager	Farming and Manufacturing, for instance coal.	Was involved in the ID cards in South Africa and the scanners.
Respondent 9	In this position, 9 months	Solutions Architect anything to do with IoT and Connectivity	Mines, Industry, Manufacturing plants and FMCG	A monitoring system in a food manufacturing plant.
Respondent 10	Two years, with 19 years overall experience	Senior Technical Architect	Mining	I was involved from an architectural perspective.
Respondent 11	Declined			
Respondent 12	Declined			

4.4 CATEGORY 2 (QUESTIONS 2–6 FROM THE INTERVIEW GUIDE)

This section focused on the extraction of guidelines from the interviewees. Important considerations that could contribute to the implementation guidelines would be identified as findings and would be discussed based on the questions bank, the intent of the question and an overview of the responses gained. The findings are obtained from the responses of the interviewees in the following sections and contributions to the implementation guidelines are in consideration throughout.

Interview Question 3A: Are there any formal guidelines used in implementing the IIoT in the industrial organisation?

These interview questions were compiled to determine if there are guidelines in existence that were not discovered by the researcher during the literature review that the respondents might have used in the implementation of the IIoT. From the interview feedback, it was notable that none of the interviewees could direct the researcher to any formal set of guidelines.

Respondent 2 commented that:

> *There are formal guidelines in terms of the communications regulator without specific reference to the IIoT guidelines as a whole.*

Respondent 3 stated that there are guidelines out there that he has seen with other IIoT implementers used on a used case basis. There is also a tendency to not share guidelines due to their competitive advantage that they need to maintain. It is also the intellectual property of some of these companies.

Respondent 4 stated that:

> *There are no formal guidelines.*

He further explained that they make use of their own set of guidelines. Respondent 9 also stated:

> *There are not any guidelines in place except for manufacture recommended specifications for the products themselves.*

In line with Respondent 4, Respondent 5 added:

> *There were no guidelines; I think we are learning as we go along. Only from a project management view, we use formal guidelines of project management according to PM BOK.*

Furthermore, Respondent 6 mentioned that they use only site-specific designs in terms of hardware. Respondent 7 mentioned that there would not be guidelines because it is a new field. Respondents 1, 8 and 10 mentioned they had no specific guidelines.

FINDING 17: No identification of formal IIoT guidelines.

Finding 17 was also evident in the literature review. The knowledge gained from the literature review was in the form of sets of recommendations and not guidelines as such.

Question 3B: In your view, if there are formal guidelines used, what should be added to these guidelines?

This question would poll the interviewee further regarding the prior Question 3A. If they answered that there are formal guidelines and they used these, the question would need to answer what the shortcomings were on these guidelines. The respondents that mentioned that there are no guidelines naturally did have nothing to add to these.

Respondent 2 mentioned:

> *The Independent Communications Authority of South Africa (ICASA) regulations need valuations of radiation patterns.*

Respondent 3 pointed out the people aspect of the IIoT solutions. He specifically mentioned the change management and adoption of these types of solutions. He believed that organisations do not adopt these IIoT solutions if there was no proper change management. Improper change management could have far-reaching effects for the organisation that invests in an opportunity and expects a return on investment. The rest of the respondents did not have anything to add because they did not identify formal guidelines.

FINDING 18: Communications regulators need to align their regulations for IIoT communications.

FINDING 19: Poor adoption of solutions if the proper change management did not occur post IIoT implementation.

It is noteworthy to mention that there is no mention of South African communications regulatory requirements within the literature readings.

Question 3C: Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?

The intention was that the interviewee offers guidelines that they used during their IIoT implementations. Offered guidelines ranged from consideration of the network and measuring devices reliability and stability. Reliability was deemed important because of the harsh mining environment wherein the architecture resides. Relevant implementation experience and a need for the right skills were also named as important considerations during the implementation of the IIoT and the maintenance after installation of the solution to the business problem, together with assessment.

Respondents further pointed out the importance of abiding by the business case and the company's non-disclosure agreements with a consideration of the site-specific constraints for the installation of sensors. Learnings could be directed on new installations from traditional implementation methods.

Respondent 3 specifically named the following guidelines:

> *Firstly, an assessment of the current business environment is needed. Compare this environment against where the business would want to position themselves in the future. They also need to be educated on what the possibilities are in terms of Industry 4.0. There needs to be a clear guideline in terms of what the business information needs are. The storage that you would need to host the data.*

The importance of a baseline assessment as described by Respondent 3 would be to assess if there is existing infrastructure in place that could be used for the new implementation. A baseline assessment would further aid at a later stage to compare the new implementation against. Respondent 4 felt the need to know the specific requirements of the customer before any commencement of even a base case assessment. Respondent 5 mentioned that new introduced equipment in an existing environment could pose challenges. Therefore, it could be beneficial to assess the compatibility of the equipment during the base case assessment. Question 3C produced findings 20–30:

FINDING 20: Before implementation, we need to understand the business/user requirements and information needs. Document business/user requirements.

FINDING 21: Compile a design document based on the user requirements before any IIoT implementation.

FINDING 22: Before implementation of the IIoT, an assessment of the current business environment is needed.

FINDING 23: A clear vision of where the business wants to position themselves in the future is required.

FINDING 24: Education of the business in terms of the possibilities of Industry 4.0 is vital.

FINDING 25: It should be considered whether there is enough data storage available for the information.

FINDING 26: Consider the challenges of new technology introductions into the existing infrastructure and environment in terms of the connectivity, security and platforms.

FINDING 27: Network needs to be reliable and stable to prevent loss in communication that could lead to production losses.

FINDING 28: The reliability and robustness of the measuring devices.

FINDING 29: The relevant skills at the different levels of architecture that are needed for IIoT implementations.

FINDING 30: It is important to abide by the organisation's rules and procedures.

Question 3D: What guidelines for an IIoT would you suggest would aid future IIoT implementations?

The intention was that the interviewee offers guidelines that they became aware of after successful completion of an IIoT solution. Question 3D was an extension of the prior question in Question 3C with the intent of further probing the interviewee on the topic of implementation guidelines. While Respondents 6, 7, 8 and 10 felt that they answered the question in Question 3C, Respondent 4 further contributed in terms of the architecture considerations and named the importance of communication and security between the different aspects of the architecture.

The other respondents cited the importance of determining if the implementation would occur in an existing or new site, while keeping in mind the constant changes in technology of the IIoT and the hardware compatibility if it is in existing sites. The importance of the IIoT aspects in terms of safety, reliability and security within the industrial organisations is regarded as one of the most important findings. Question 3D discovered the following findings 31–40.

FINDING 31: Consider the availability of networks and if it is possible to use these networks.

FINDING 32: Consider the platforms needed for the IIoT solution and if it would easily integrate with existing platforms.

FINDING 33: There is a need for a secure communications protocol and data security.

FINDING 34: Determine if the implementation is a new (greenfields) installation or an existing (brownfields) installation.

FINDING 35: Industrial communication protocols need to be adapted to the low-power extensive area network technology if they have not been adapted yet. Changes within the IIoT hardware and platforms should not hinder the implementation of an IIoT solution.

FINDING 36: It should be determined if the new hardware is compatible with the existing infrastructure**.**

FINDING 37: A clear understanding is necessary of the technology that needs to be implemented.

FINDING 38: IIoT security is imperative in an industrial environment.

FINDING 39: Pre-implementation to define the key performance indicators that need to be achieved is important.

FINDING 40: Monitoring of the IIoT deliverables and if they are achieved should be determined.

Question 4 A: In your view, what industries could benefit from IIoT implementations?

This question defines the landscape that would benefit from IIoT implementations. It might be possible that specific industries do not need any such implementations. Of the respondents, 80% stated that industrial industries would benefit from IIoT implementations while Respondent 6 felt that it would mostly benefit remote areas. Respondent 2 thought it is more applicable in situations where industries manufacture something. This led to Finding 41 that mentions IIoT has the potential to benefit industrial organisations.

FINDING 41: IIoT has the potential to benefit industrial organisations; therefore, successful implementations are of the essence.

Question 4 B: In your opinion, what needs to be in place for an IIoT implementation?

In this question, required items in terms of IIoT implementations are identified and defined. Without these pre-requirements, an IIoT implementation would be unsuccessful.

Prerequisites ranged from the needed IIoT architecture (importance of robustness) with the support of the right implementations partner. The implementations partner would need a high level of skills. There is also a concern raised in terms of the willingness of adoption by the business. The IIoT implementation needs to be supported by a detailed architecture design. This question produced the follow findings:

FINDING 42: IIoT hardware should be able to handle the environments in which they function (robustness).

FINDING 43: If no reliable connectivity to cloud is in place, analytics needs to be handled on-site rather than putting the business at risk.

FINDING 44: Needed IIoT infrastructure includes **IIoT** devices, networks, network coverage and platform for cloud infrastructure.

FINDING 45: Persons with the appropriate experience in delivering IIoT solutions in the specific industry are crucial.

FINDING 46: Detailed architecture is beneficial when the implementation of IIoT solutions commences.

FINDING 47: It is vital to understand the requirement of the business.

FINDING 48: Relevant people skills are needed during the implementation and after that for maintenance.

FINDING 49: Hardware redundancy is deemed an important consideration; where there is an interruption of an important manufacturing process, a backup system could assist during the failure of the primary system to prevent the interruption.

FINDING 50: Robustness of IIoT devices is an important consideration due to the harsh environment (extreme heat cold, dust or moisture ingress) where these devices function.

FINDING 51: Vision of the end state after the IIoT implementation.

FINDING 52: Relevant procedures should be in place to manage the IIoT infrastructure and environment.

FINDING 53: The correct technology is required for the application of the solution that needs to solve the business problem.

FINDING 54: Proper governance for the IIoT environment should be in place.

Question 4C: Is there any relevance if the organisation is already in existence with connected sensors or if it is a new organisation embarking on IIoT?

Question 4C aimed to understand the implications that existing architecture or the legacy systems could have on the implementation. Of the respondents, 70% agreed that existing infrastructure would have an effect on new implementations. Considerations would be if the existing architecture or portions thereof could be used or integrated.

FINDING 55: Challenges with existing infrastructure and platforms exist when implementing IIoT.

FINDING 56: IIoT can complement existing technology investments.

FINDING 57: Greenfields organisations have no challenges in terms of legacy systems

Question 5A: Please draw the architecture needed for an IIoT implementation.

Question 5A presents the key considerations and components of the architecture needed for IIoT implementation. Valuable insights pertaining to the architecture were obtained from this particular question and led to the construction of a new IIoT architecture view as per Figure 6. Table 5 contains a summary of responses in terms of architecture.

FINDING 58: At the bottom layer of the IIoT architecture, measuring sensors – also called things or devices – acquire the data on the equipment that needs to be monitored, analysed, reported on or visualised.

FINDING 59: Data collection layer with the software to enable the collection of data. These services could exist in the cloud or at the business site.

FINDING 60: Storage of data could be local on hardware on-site or stored in cloud storage. The storage of data should be able to handle structured and unstructured data.

FINDING 61: Software platforms that have the ability (depending on the business need) to analyse, visualise, recognise patterns in the data and report on the data. These services can find correlations to other data from other sources. These services could exist in the cloud or at the business site.

FINDING 62: Throughout all the layers network connectivity is necessary for the flow of the data.

FINDING 63: Security of the data is essential in all the different layers with the help of gateway hardware.

Table 5: Summary of responses

Participants	IIoT Architecture response from respondents	Key highlights in terms of needed architecture
Respondent 1	*Connected devices, network, on-premise and offsite services for instance cloud services.*	Devices, network connectivity, services (on-premise or offsite)
Respondent 2	*Technologies that companies can use to increase efficiencies in terms of manufacturing, supply chain and cost savings.*	Technology
Respondent 3	*Connected things in the manufacturing aspects of the business that help with the digitisation of the business. The aim to create even more useful data than the past.*	Things, connection, data
Respondent 4	*Sensors at the bottom layer in line with what needs to be measured, for example pressure and temperature. An internal or external network. An edge device or cloud platform. A storage device to store the data. Using cloud services for analytics. Also, ways to visualise it to make some intelligent business decision. Drawing correlations between different data sets and data points across multiple locations.*	Measuring sensors, edge device, data storage, cloud platform for analytics, visualisation and pattern recognition
Respondent 5	*Using modern media, modern technology in the industrial environment to streamline processes to improve efficiencies, control plant better, which will go from your plant to your phone at home, so completely integrated solutions.*	Modern technology

Respondent 6	*Connecting anything anywhere and obtaining and moving data to a centralised place for analysis and visualisation of the data.* *A platform where the data can be hosted, data layer, could also be in the cloud: network connectivity and the measuring device.*	Connected things with network connectivity, centralised storage of data for interrogation, analysis and visualisation
Respondent 7	*Separate devices are interacting with each other and integrating with internet that host cloud services.* *Devices, network and security.*	*Devices, network and security*
Respondent 8	Measuring equipment that gathers the data for information, connected to a physical network for communication to data storage or cloud services.	Measuring equipment, data storage, network, cloud services
Respondent 9	*Sensors at the edge of the field, a gateway to the cloud and processing in the cloud. In the cloud platform, you have dashboards and analytics etc.*	Field or edge sensors, a gateway to the cloud and processing of data in the cloud in terms of visualisation and analytics
Respondent 10	*Connected things that could generate data.* *Data collection zone to pull in or push the needed information. Data storage zone for the storage of structured and unstructured data.*	Data collection and data storage that could handle structured and unstructured data

Question 5B: In your opinion, what are some of the critical considerations for any industrial organisation embarking on an IIoT implementation?

Question 5 defines other aspects that are not evident from the architecture and would need consideration. Respondent 4 mentioned the importance of the need for any proposed IIoT solution and stated that:

> *There should be a problem that needs to be solved that would create business value and a return on investment.*

Respondent 2 enquired what the cost would entail to solve the problem. An important aspect is that there needs to be a return on the investment in the technology.

Respondent 5 also raised the cost of the solution as a consideration and added that a skilled implementation team is necessary. Training of the personnel that need to support the implementation must take place. Most of the other respondents agreed that the value that an IIoT implementation could have is important and to ensure that there is value. Related findings that were identified from the interviews in terms of this question include the following:

FINDING 64: Clear business case that justifies the implementation of the technology.

FINDING 65: Outcome of the implementation is known.

FINDING 66: Robustness of the equipment is important.

FINDING 67: Change management should be evident during implementation.

FINDING 68: There should be justification of the outcomes against the cost.

FINDING 69: System uptime (reliability is critical).

FINDING 70: Skilled implementations team is vital.

Question 6A: Please name a given case where you implemented an IIoT and what you would have done differently given the same situation.

The aim of Question 6A was to build credibility in terms of the interviewees and their involvement in IIoT. The researcher reasoned that for a respondent to be able to comment on implementation guidelines of IIoT, they should have been involved in implementations. All the interviewees mentioned cases that they were part of that could verify their involvement in IIoT implementations. Further findings arose within the discussion of Question 6A as the interviewees remembered relevant data because they discussed known implementation cases personal to them. These findings are as follows:

FINDING 71: The availability of the chosen devices in the measuring level and the reliability thereof.

FINDING 72: A complete understanding of the customers' needs and their expectations.

FINDING 73: Challenges in terms of people skills regarding IIoT.

FINDING 74: Challenges in terms of communication protocols in use.

FINDING 75: Challenges in terms of existing legacy systems and hardware.

FINDING 76: Network connectivity challenges**.**

Question 6 B: What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?

The interviewer reasoned that lessons learned would aid future organisations to prevent pitfalls in IIoT implementations. In regard to lessons learned, Respondent 4 stated that he learned a valuable lesson:

> *To methodically lay out the whole process of where the data originates. Where the data should go and what the purpose of the data is together with the need for it before any implementation starts. Implementation of a large scale would mean a few months*

> *to install the necessary units — the maintenance of the whole IIoT solution after the time is also a consideration.*

In support of this idea, Respondent 5 iterated the need for a clear architectural design for the IIoT implementation. Some of the other respondents mentioned alternative lessons learned as captured in the following findings:

FINDING 77: Data flow mapping of where the data originates from to where it is needed.

FINDING 78: Time needed for a large-scale IIoT implementation.

FINDING 79: The personnel needed after the installation to maintain the implementation.

FINDING 80: The robustness of the IIoT equipment.

FINDING 81: A complete and precise design for the IIoT before implementation.

FINDING 82: Small achievable project milestones for an IIoT implementation.

FINDING 83: Clients need to be given preference in the IIoT planning.

Question 6C: What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?

The interviewees offered IIoT implementation considerations during this question that ranged from liaising with a competent service provider to implement successful solutions, knowing the end state, advising during the physical implementation and researching the technologies available. Respondent 2 recommended before any implementation:

> *A problem that exists could be solved by an existing solution that is easily implementable to prove business value, have the budget upfront and understand the timelines for the resolution. Also, who will provide it?*

Respondent 8 mentioned that for the post-implementation, an important consideration is if the IIoT implementation was successful and the industrial organisation could produce more, could they offset the additional product?

FINDING 84: An agile approach to implementation could be more favourable because of the rapidly changing environment.

FINDING 85: Post implementation, ensure that the additional product that arises from a successful IIoT implementation could be sold and that the logistics are in place.

FINDING 86: During the planning phase, research is needed in terms of the IIoT solution.

Question 6D: What would you describe as the advantages of an IIoT implementation?

Question 6D is the last of the questions to extract guidelines. The researcher needed to confirm that there are advantages in IIoT implementations. If there are advantages, it would mean that there is real value in successful IIoT implementations and that would mean that there is a need for guidelines to ensure the successful implementation of IIoT. Finding 87 sums up the shared ideas from the respondents and is also the last of the findings.

FINDING 87: Advantages include gaining market share, increased efficiencies, insight into business operations and insight into different aspects of the organisation.

A total of 87 findings were discovered throughout the use of the interview questionnaire during the interview sessions. The findings are categorised and arranged according to related themes in the next section.

4.5 CATEGORIES AND THEMES

The developed themes were derived from the findings attained during the interview questions. These themes were then categorised, as seen in Table 6.

Table 6: Discovered categories related to themes

Categories	Themes
IIoT architecture	Hardware
	Software
	Security
	Communication
Implementation	Pre-implementation
	Implementation
	Post-implementation
IIoT knowledge	Training and awareness
	Challenges
	Advantages

4.5.1 IIOT ARCHITECTURE AS A CATEGORY

IIoT architecture refers to the specific guidelines needed to address the infrastructure required for IIoT implementation. These infrastructures were identified by the interviewees and from literature as hardware, software, security and the communication of the components of the IIoT. The themes that are categorised within the IIoT architecture are hardware, software, security and communication.

4.5.2 IMPLEMENTATION AS A CATEGORY

This category refers to the planning and consideration before, during and after implementation. Pre-implementation refers to the steps and actions needed before the actual implementation of an IIoT can be attempted and would form part of the planning of the IIoT implementation. Some aspects that need to be planned are not known. If it were known, projects would not have had the overruns that are currently the norm. Planning in terms of reliability of the solution should also be addressed.

The during-implementation or implementation theme identified from the findings helps in understanding the different aspects that need to be addressed during the implementation that would help to ensure the success of implementation. The post-implementation findings aim to address the maintenance and upkeep of the investment in the IIoT.

4.5.3 IIOT KNOWLEDGE AS A CATEGORY

From training and awareness as a finding from the interviewees and the newness of the IIoT implementations, it can be seen that there is still work to be done in terms of educating businesses to understand the value of IIoT. Lack of understanding creates challenges in IIoT considerations. A lack of awareness also hinders the industrial organisations from seeing the advantages and challenges that form part of the IIoT implementation. Related themes that are part of IIoT knowledge as a Category, are the Training and Awareness, Challenges and advantages.

4.6 SUMMARY OF FINDINGS GROUPED ACCORDING TO CATEGORIES AND THEMES

By asking a total of 18 interview questions to 10 respondents to extract IIoT implementation guidelines, the summary of findings is displayed in Table 7. These findings are grouped according to the identified themes and categories.

Table 7: Summary of findings from the interviewees

Finding No.	Finding Summary	Theme
	Category: Architecture	
44	Needed IIoT infrastructure includes IIoT devices, networks, network coverage and platform for cloud infrastructure.	Architecture
	Hardware	
11	High resilience to failure of hardware against environmental factors.	Hardware
25	Consider if there is enough data storage available for the information.	Hardware
28	The reliability and robustness of the measuring devices.	Hardware
42	IIoT hardware should be able to handle the environments in which they function (robustness).	Hardware
49	Hardware redundancy is deemed an important consideration; where there is an interruption of an important manufacturing process, a backup system could assist during the failure of the primary system to prevent the interruption.	Hardware
50	Robustness of IIoT devices.	Hardware

58	At the bottom layer of the IIoT architecture, measuring sensors – also called things or devices – acquire the data on the equipment that needs to be monitored, analysed, reported, on or visualised.	Hardware
59	Data collection layer with the software to enable the collection of data. These services could exist in the cloud or at the business site.	Hardware
60	Storage of data: it could be local on hardware on-site or stored in cloud storage. The storage of data should be able to handle structured and unstructured data.	Hardware
66	Robustness of the equipment.	Hardware
71	The availability of the chosen devices in the measuring level.	Hardware
Software		
10	The level of intelligence needed at the measuring device level and the ability to handle automation on the lower level.	Software
15	The industrial systems should be easily reprogrammable to support changing processes.	Software
32	Consider the platforms needed for the IIoT solution and if it would easily integrate with existing platforms.	Software
61	Software platforms that have the ability (depending on the business need) to analyse, visualise, recognise patterns in the data and report on the data. These services can find correlations to other data from other sources. These services could exist in the cloud or at the business site.	Software

69	System uptime (reliability is critical).	Software

Security		
12	A high level of security to prevent impacts on production and safety.	Security
33	There is a need for a secure communications protocol and data security.	Security
38	IIoT security is imperative in an industrial environment.	Security
63	Security of the data in all the different layers with the help of gateway hardware.	Security
Communication		
18	Communications regulators need to align their regulations for IIoT communications.	Communication
27	Network needs to be reliable and stable to prevent loss in communication that could lead to production losses	Communication
31	Consider the availability of networks and if it is possible to use these networks.	Communication

43	If no reliable connectivity to cloud is in place, analytics needs to be handled on-site rather than putting the business at risk.	Communication
35	Industrial communication protocols need to be adapted to the low-power extensive area network technology if they are not yet changed within the IIoT hardware and platforms should not hinder the implementation of an IIoT solution.	Communication
62	Throughout all the layers, network connectivity is needed for the flow of the data.	Communication

Finding No	Finding Summary	Theme
	Category: Implementation	
	Pre-Planning	
1	Before any implementation commences, the creation of a clear baseline is needed that considers the current data collection methods, the existing infrastructure vs the needed IIoT architecture.	Pre-Planning
2	If there are cloud services used, the presence of data analytics, the relevant needed reports for decisions and the level of automation.	Pre-Planning
3	Consider the business demographics.	Pre-Planning

4	Thorough planning is needed before any IIoT implementation can commence.	Pre-Planning
5	Before implementation, consolidation of different source data is necessary.	Pre-Planning
8	Scalability of IIoT devices when they grow into the thousands.	Pre-Planning
13	Compatibility of newer equipment and systems within IIoT to function with existing or legacy systems.	Pre-Planning
14	IIoT solution should have a very high level of precision and accuracy.	Pre-Planning
16	IIoT should have a low latency because of the high-speed production systems in use.	Pre-Planning
17	No identification of formal IIoT guidelines.	Pre-Planning
20	Before implementation, we need to understand the business/user requirements and information needs. Document business/user requirements.	Pre-Planning
21	Compile a design document based on the user requirements before any IIoT implementation.	Pre-Planning
22	Before implementation of the IIoT, an assessment of the current business environment is needed.	Pre-Planning
23	A clear vision is needed of where the business wants to position themselves in the future.	Pre-Planning

37	A clear understanding is necessary of the technology that needs to be implemented.	Pre-Planning
39	Pre-implementation to define the key performance indicators that need to be achieved.	Pre-Planning
47	Understand the requirement of the business.	Pre-Planning
51	A vision of the end state after the IIoT implementation is needed.	Pre-Planning
53	The correct technology for the application.	Pre-Planning
64	A clear business case that justifies the implementation of the technology.	Pre-Planning
68	Justification of the outcomes against the cost.	Pre-Planning
72	A complete understanding of the customer's needs and expectations.	Pre-Planning
77	Data flow mapping of where the data originates from to where it is needed.	Pre-Planning
78	The time needed for a large-scale IIoT implementation.	Pre-Planning
81	A complete and transparent design for the IIoT before implementation.	Pre-Planning

83	Clients' needs are the basis for IIoT Planning.	Pre-Planning
9	Serviceability of the industrial equipment. Equipment should be easily maintainable.	Pre-Planning
79	The personnel required after the installation to maintain the IIoT implementation.	Pre-Planning
85	Post implementation, ensure that the additional product produced from a successful IIoT implementation could be sold and that the logistics are in place.	Pre-Planning
34	Determine if the implementation is a new (greenfields) installation or an existing (brownfields) installation.	Pre-Planning
36	Determine if the new hardware is compatible with the existing infrastructure**.**	Pre-Planning
	Execution	
46	Detailed architecture is beneficial when the implementation of IIoT solutions commences.	Execution
67	Change management during execution.	Execution
70	Skilled implementations team.	Execution
82	Small achievable project milestones for an IIoT implementation.	Execution

84	An agile approach to implementation could be more favourable because of the rapidly changing environment.	Execution
Post-Implementation		
40	Review of the IIoT deliverables and if they are completed.	Post Implementation
52	Relevant procedures to manage the IIoT infrastructure and environment.	Post Implementation
54	Proper governance for the IIoT environment.	Post Implementation
65	The outcome of the application is known.	Post Implementation
Category: IIoT Knowledge		
Training and awareness		
7	Lack of in-house skills for IIoT implementations and the maintenance thereof after.	Training and awareness
24	Education of the company in terms of the possibilities of Industry 4.0	Training and awareness
29	The relevant skills at the different levels of architecture that are needed for IIoT implementations.	Training and awareness

30	Essential to abide by the organisation's rules and procedures.	Training and awareness
48	Relevant people skills are needed during the implementation and after that for maintenance.	Training and awareness
	Challenges	
6	Management of cultural and organisational changes.	Challenges
19	Poor adoption of solutions if the proper change management did not occur post IIoT implementation.	Challenges
26	Consider the challenges of new technology introductions into the existing infrastructure and environment in terms of the connectivity, security and platforms.	Challenges
55	Challenges with existing infrastructure and platforms exist when implementing IIoT.	Challenges
57	Greenfields organisations have no challenges in terms of legacy systems.	Challenges
73	Challenges in terms of people skills in terms of IIoT.	Challenges
74	Challenges in terms of communication protocols in use.	Challenges
75	Challenges in terms of existing legacy systems and hardware.	Challenges

76	Network connectivity challenges**.**	Challenges
	Advantages	
41	IIoT has the potential to benefit industrial organisations.	Advantages
56	IIoT can complement existing technology investments.	Advantages
87	Advantages include gaining market share, increased efficiencies, insight into business operations and insight into different aspects of the organisation.	Advantages

4.7 CHAPTER SUMMARY

During Chapter 4, the resolve was to analyse the data collected through the semi-structured interviews and literature research. These semi-structured interviews were conducted based on interview questions that would answer the main research questions and research sub-questions.

An analysis concluded in a total of 87 identified findings from the interviews. These 87 findings resulted in discovering ten themes and three categories.

The ten identified themes were:

1. Hardware
2. Security
3. Software
4. Communication
5. Pre-implementation
6. During implementation
7. Post-implementation
8. Training and awareness
9. Challenges
10. Advantages

And the three categories were:

1. IIoT architecture
2. Implementations
3. IIoT knowledge

Chapter 5 will further discuss these findings and themes concerning the research questions in an attempt to extract the guidelines from the research.

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CHAPTER 5: DISCUSSION

5.1 INTRODUCTION

Chapter 5 discusses the research findings obtained in Chapter 4, according to the discovered themes. From this discussion, answers to the research questions stated in Chapter 1 will be offered together with the presentation of the ten developed themes.

5.2 PROBLEM STATEMENT AND RESEARCH QUESTIONS

5.2.1 PROBLEM STATEMENT

Implementation of the IIoT realised only recently in the industrial sectors because of maturity and affordability. The problem due to the hardware that only recently became affordable and the development of the analytic tools on an application layer was that only a few IIoT implementations had been completed in the industrial environment. Further to this, literature or guidelines that could assist companies in implementing an industrial IIoT solution are few.

If relevant literature or guidelines in the implementation of IIoT existed on this topic, it would enable more industries to implement and leverage the advantages of IIoT in a shorter period

too. The benefit of clear guidelines in the industrial sector would mean that implementations could realise some objectives of improvement on the safety and efficiency of an IIoT solution faster. Since there are no guidelines, it creates a gap in the literature and motivates this research.

5.2.2 MAIN RESEARCH QUESTION:

What is a list of guidelines that can be used for IIoT implementations?

5.2.3 SUB-QUESTION TO THE MAIN QUESTIONS:

- What guidelines are pertinent in the existing literature?
- What are the critical considerations for any organisation planning to implement the IIoT to gain the advantages?
- What are the lessons learned from other implementations in the industrial organisations that could assist new implementations?

5.3 FINDINGS

A set of interview questions was compiled to specifically extract answers from the interviewees where there were no data available within the literature. The answers obtained from the literature and interviews are named findings. A total of 87 findings were identified and related to the research questions as follows.

Table 8: Relation to research questions

Research questions	**Findings No.**
Main research question	
What is a list of guidelines that can be used for IIoT implementations?	Complete list of findings has relevance
Research sub-question	
What guidelines are pertinent in the existing literature?	1–16
What are the critical considerations for any organisation planning to implement the IIoT to gain the advantages?	41–70
What are the lessons learned from other implementations in industrial organisations that could assist new implementations?	71–86

5.4 THEMES AND CATEGORIES DEVELOPED

From the research findings, the identification of three categories and the development of ten themes transpired. The three categories with their respective themes about the implementation of IIoT were:

- Architecture with the themes in consideration of hardware, software, security and communication;
- Implementation with the themes of pre-implementation, during implementation and post-implementation; and
- IIoT knowledge with the themes of training and awareness, challenges and advantages.

A discussion follows on the categories and the themes within the categories. Following the explanation of the categories and themes, findings that complement or enforce the same ideas would be merged, and the data would be presented in an overall guidelines list that would answer the main research question.

5.4.1 IIOT ARCHITECTURE

IIoT architecture is needed for any implementation of the IIoT. Without the hardware, software, security and communications, it would not be possible to implement an IIoT solution. Figure 8 presents IIoT architecture as discovered from the literature and interviews.

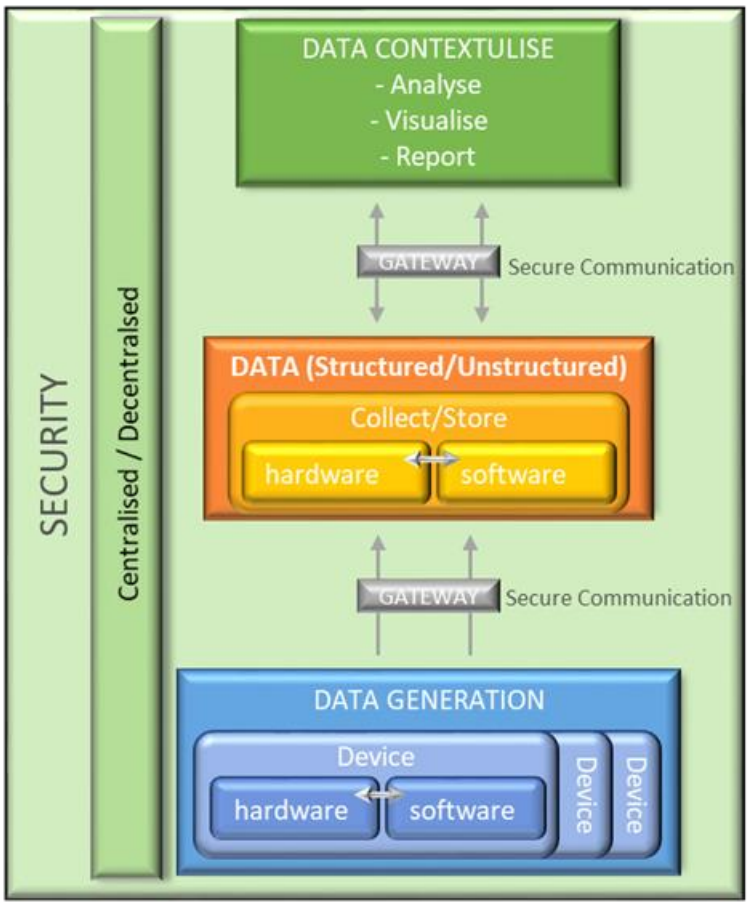

Figure 8: IIoT architecture as discovered from the literature and interviews

Concerning Figure 8, the data generation layer contains the measuring devices, also called things in the IIoT, which measure different parameters. These devices consist of hardware and software components. There could be M2M communication within the device level, and the different communication protocols are of relevance. Automation could also be within this level on IIoT. Security on this level is necessary to ensure data integrity and to prevent stoppages in the production environment. Measuring devices could communicate wirelessly with the data collect and store layer, and a high level of robustness is needed for the continuous functioning of the hardware. Data flows from this lower level to the data collect and store layer.

The data collect and store layer identified from the findings also consists of a hardware and software component. The hardware could be locally on the premises or offsite cloud services could be used. The hardware is used in conjunction with software to collect and store data. There is also a high level of security needed in this layer.

Lastly, the data contextualise layer – be it on-site, offsite or in the cloud – presents the data in a way that the business could make sense of the data. Data becomes information within this layer and could be visually presented in the form of a report or a SCADA display. Analytics could also be done within this layer, as well as machine learning and pattern recognition.

Hardware

A critical consideration in the selection of the hardware in an industrial environment is hardware that can withstand the harsh production conditions. For instance, in a mining environment, the environmental factors in the mining area could be extreme heat, dust, mechanical forces and rain. For any hardware to keep on functioning, they must be able to withstand these elements. It is also crucial for the hardware to keep functioning within an industrial environment. If the hardware were to fail, it could mean safety incidents or production losses. Hardware should be highly reliable, robust, possess a high resilience to failure and be available whenever the process needs their functioning. Respondents even mentioned that in certain areas hardware, redundancy is in place to ensure that a particular process is backed up with a set of standby equipment in the case of hardware failure.

According to the responders, the following hardware needs to be in place for any IIoT implementation to be successful:

- IIoT devices
  Considering the scalability of IIoT devices when they grow into the thousands was also cited as a consideration before the implementation of the IIoT hardware devices. The respondents questioned whether there would be enough technical resources to install

the magnitude of sensors required in installation going into the thousands of sensors. They also questioned if the supporting hardware that hosts the platforms would be able to process the amount of data.

- Networks and network coverage for the communication between the different layers.
- A hardware platform host could be in the cloud or on-premise. This hardware platform hosts the data storage, collection and contextualisation of the data. From the findings, it should be noted that if there is no reliable connectivity to the cloud available, the storage, collection and contextualisation of the data needs to be handled on-site. This is in order not to expose the business to risk if there are communication interruptions to the cloud that could affect a critical process. An example could be when a manufacturing process that moulds plastic loses connectivity to the cloud and the process halts. It should be considered whether there is enough hardware space allocated for the storage of the information and the hosting of the services.
- Gateway devices that aid security solutions.

Software

Another guideline to consider according to the findings is the software platforms needed for the IIoT solution and whether these platforms could easily integrate with existing platforms. In this regard, Karschnia (2015) mentions that these “things” interconnect on- and offsite to software platforms to enable remote control, monitoring and asset management using either a dedicated expert team or specialised data analysis software connected to big data lakes to add value. From the responders, it is essential to do a consolidation of different source data before implementation.

Within the IIoT, intelligence could move down to the instrument level. If this is the case, the level of intelligence needed at the measuring device level and the ability to handle automation on the lower level are an important consideration and the software should accommodate the functionality.

The industrial systems should be easily and frequently reprogrammable to support changing processes. Within complex industrial processes, process improvements are regularly attempted. The systems that support these processes therefore need to be easily adaptable.

A consideration would be in terms of the platforms needed for the IIoT solution and if it would easily integrate with existing platforms. Platforms are relevant where the IIoT implementation is on an existing site.

The software should be highly reliable, robust, possess a high resilience to failure and be available whenever the process needs their functioning. System uptime is critical in industrial processes.

Security

There is a need for a secure communications protocol and data security as a guideline for implementation. Security is especially relevant in the IIoT environment that is reliant on the safety, security and reliability of their operations. IIoT has a higher level of security due to the critical nature in a disruption of a high-volume manufacturing process or the takedown of the electrical grid that could have an economic impact (Chan, 2017). Raynor and Cotteleer (2015) posit that security challenges need to be kept in mind before any implementations commence. Holdowsky et al. (2019) also point to the security requirements in the communications protocol that could have an impact on the safety, security and reliability within an IIoT implementation. Soldatos (2016) agrees that one of the critical concerns is the security aspect around the transmittal of information within the IIoT.

Communication

Raynor and Cotteleer (2015) define communication as a state for the transmission of information from one place to another and then aggregating or gathering the data from the different sources and times. The respondents identified availability of networks and whether it is possible to use these networks as a concern. The industrial communication protocols should be adapted to the low-power extensive area network technology. The existing IIoT

hardware and software platforms should not hinder the implementation of an IIoT solution. Existing networks should possibly be considered for IIoT implementation. Legislation within South Africa needs to accommodate the communications between devices. Respondent 3 referred to legislation and said that: “Communications regulators need to align their regulations with regard to IIoT communications.” Without reliable communication between the different levels within the IIoT, no successful implementation could take place.

The communications should be highly reliable, robust, possess a high resilience to failure and be available whenever the process needs their functioning. Communications uptime is critical in the industrial processes, and the solution should have a low latency because of the high-speed production systems in use.

5.4.2 IMPLEMENTATION

Pre-planning

Before implementation of any IIoT solution, a need or a problem to be solved must exist. A clear understanding of the business/user requirements and information is imperative. The business need should be supported by a clear business case that justifies the implementation of the technology and ensures that there is a return on the investment in the technology. This business requirement should be documented to ensure that the IIoT solution would support the need.

A detailed design document, based on the user requirements, should then be created before any IIoT implementation. This detailed design could direct the implementation of the IIoT solution. This design needs to consider the key performance indicators that need to be achieved and the available technology must be considered. The detailed design should also determine if the implementation is a new (greenfields) installation or an existing (brownfields) installation. It should be determined if the existing infrastructure and architecture are compatible with the existing infrastructure. A decision on the technology should be made. This selection of technology would need to consider the ease of

serviceability and maintainability of the technology and upgrade paths during the life cycle. The business should have a clear vision of how the organisation would function after the implementation, and what the expectation is of the implementation. In terms of personnel, the planning for skilled personnel is vital after the installation to maintain the IIoT implementation. Consideration should be given, if the solution implementation is a success, to the advantages of the IIoT implementation. With an increase in production, it should be considered whether the additional product produced from a successful IIoT implementation is sellable and whether there is there a market. It should be determined whether the organisation's logistics can accommodate the increase in the product**.**

Blanchette (2016) emphasised the importance of thorough planning before any IIoT implementation commences. Planning was also cited as an essential factor before any IIoT implementations could commence as there is no identification of formal IIoT guidelines.

In this regard, Lueth (2016) states that IoT projects take over 18 months longer than what had been initially planned. At the base of thorough planning, Blanchette (2016) emphasises the creation of a baseline (as-is) situation of the organisation. The responders mentioned that the baseline should consider:

- if there are cloud services used, the presence of data analytics, the relevant needed reports for decisions and the level of automation;
- the business demographics;
- scalability of IIoT devices when they grow into the thousands;
- compatibility of newer equipment and systems within IIoT to function with existing or legacy systems;
- consolidation of different source data, before implementation;
- a clear vision of where the business wants to position themselves in the future;
- a clear understanding of the technology that needs to be implemented; and
- data flow mapping of where the data originates from to where it is needed.

Execution

Detailed architecture is needed when the implementation of IIoT solutions commences and needs to contain decisions on the selected technology. Change management during execution is critical to ensure that the solution is adopted and the benefits are realised. During the execution of the IIoT implementation, it is vital to use a skilled implementations team. Respondents' recommendations for during the implementation of IIoT solutions were that small achievable project milestones should be planned and pursued to ensure the success of IIoT implementation. The time needed for a large-scale IIoT implementation should be kept in mind.

Post-implementation

A review should take place of the IIoT deliverables and whether these deliverables have been met. Relevant procedures and the necessary governance should be introduced to manage the IIoT infrastructure and environment.

5.4.3 IIOT KNOWLEDGE

Training and awareness

The findings indicated that there is a lack of in-house skills for IIoT implementations and the maintenance thereof afterwards. The lack of in-house skills for IIoT implementations poses a challenge for the organisation. There is also a lack of relevant skills at different levels of IIoT architecture. There is room for improvement in the education of companies in terms of the possibilities of Industry 4.0. Relevant people skills are needed during the implementation and after that for maintenance.

There is a need for the creation of awareness within the organisation on the management of cultural and organisational changes needed coupled with the implementation of the IIoT. These cultural and organisational changes would be a leading factor for the poor adoption of the implemented solutions if the proper change management does not occur.

Awareness and training should occur to cater for any lack of in-house skills for IIoT implementations and the maintenance thereof after the implementation has taken place. Proper maintenance would mean that competent personnel are needed after the installation to maintain the implementation. In this regard, education for the business in terms of the possibilities of Industry 4.0 is needed for the organisation to realise the potential value-adding business cases.

Challenges

The challenges that implementations of IIoT should be aware of according to the findings are that there need to be people with the appropriate experience in delivering IIoT solutions in the specific industry in consideration. Another challenge could emerge with existing infrastructure and platforms when attempting an IIoT implementation. Further challenges include:

- management of cultural and organisational changes;
- poor adoption of solutions if the proper change management did not occur post IIoT implementation;
- new technology introductions into the existing infrastructure and environment in terms of the connectivity, security and platforms;
- existing infrastructure and platforms exist when implementing IIoT;
- greenfields organisations have no challenges in terms of legacy systems;
- people skills in terms of IIoT;
- existing communication protocols in use;
- existing systems and hardware that are in use; and
- network connectivity issues**.**

Advantages

The advantages described are that the IIoT has the potential to benefit industrial organisations and can complement existing technology investments. Typical advantages

include gaining market share, increased efficiencies, insight into business operations and insight into different aspects of the organisation.

5.5 IMPLEMENTATION GUIDELINES

In answer to the main research question of "What is a list of guidelines that can be used for IIoT implementations?", the researcher considered the research sub-questions, the findings and themes and categories in compiling the following list of guidelines in conclusion to the study. These guidelines are based on the literature review and the interviews.

Implementation guidelines for IIoT

1) There must be a clear business case or a problem to be solved by the IIoT implementation.
   a) The business case should have clear deliverables and an end-state vision.
   b) There should be a return on investment, be it in terms of monetary value or safety improvements to be a justification of the outcomes against the cost.
   c) The business case or requirement needs to be well documented to direct the IIoT implementation and should contain the key performance indicators.

2) The business requirement should then be supported by a detailed design of the proposed IIoT implementation. The architecture of IIoT, as discovered from the literature and interviews, could be used as a basis for discussion on the components that need to be considered in compiling the detailed design.

   Thorough planning is needed before any IIoT implementation could commence.

   a) In terms of the business context, the following will assist in planning:
      - Is this a completely new IIoT implementation (greenfields) or an implementation within an existing business (brownfields)?
      - What elements of the architecture exist within the brownfields organisation?

- In brownfields, can any of these elements be incorporated and integrated with the newer technologies that will be introduced in the IIoT implementation?
- Brownfields implementations should be aware of the integration challenges with existing equipment.
- Identify the challenges that existing infrastructure and platforms could have on a planned implementation concerning greenfields and brownfields organisations.
- Are there any automation technologies or any automated processes?
- Data flow mapping should be done of where the data originates from to where it is needed.
- Post implementation, ensure that the additional product produced from a successful IIoT implementation could be sold and that the logistics are in place.

b) In terms of centralisation or decentralisation of business:

- What do the demographics of the business entail? Is it centralised, decentralised or a hybrid of these two?
- Consider the location of the business services and the technology that supports these services if it is either off-premise (cloud services) or on the premises or a hybrid thereof.
- The software platform exists in the cloud, and there should be reliable communication between the organisation and the cloud service provider in order not to put the business at risk. Another consideration is the number of data points that the platform would need to handle.

c) In terms of technology:

- Selection of the technology to be implemented based on availability and maintainability in future.
- A clear understanding of the technology needs to be implemented.

- The hardware and software should be highly reliable, robust, possess a high resilience to failure and be available whenever the process needs their functioning.
- Future availability of the parts within the IIoT, for instance the sensors, hardware and software upgrades, if there is any failure of any of these parts.
- Does the possibility exist to scale the implementation of IIoT devices into the thousands if needed? Will the hardware be able to support it?
- System redundancy if there is a failure – what backup could provide the services?

d) In terms of communication:

- Network coverage, maintainability, the speed and the amount of data for transmission should be considered.
- Communications protocol for the transmittal of data should be implemented.
- Communication networks are needed between the different architecture levels of data generation, the collection and storage of data and the data contextualisation layer.
- Network reliability and stability are vital.
- Communications regulators need to align their regulations for IIoT communications.
- The availability of networks and if it is possible to use these networks should be kept in mind.
- If no reliable connectivity to the cloud is in place, analytics needs to be handled on-site rather than putting the business at risk.
- Industrial communication protocols need to be adapted to the low-power extensive area network technology if they have not yet been adapted.
- Changes within the IIoT hardware and platforms should not hinder the implementation of an IIoT solution.

e) In terms of security:

- The architecture should possess a secure communications protocol for the transmittal of data.
- There needs to be data security within the IIoT implementation.
- The necessary security in terms of protocols should be in place.
- A high level of security is needed to prevent impacts on production and safety.
- The security of the data in all the different layers should be considered.
- The use of gateway hardware in the different layers should also be considered.

f) In terms of the software:

- Choose and define the software platforms needed for the IIoT solution.
- Platforms should easily integrate with existing platforms on-site and to cloud services, for instance big data analytics. The same applies for remote control, monitoring and asset management.

g) In terms of the hardware:

- The needed hardware should consider the scalability of the IIoT implementation.
- The hardware in terms of the IIoT devices that gather the data (measuring points) should be determined. Their application and physical location to determine their robustness and reliability should be considered as well as the lifetime needed on these sensors and their maintainability.
- The hosting hardware of the software platforms.

h) Regarding the data generation layer:

- The level of intelligence needed at the measuring device level and the ability to handle automation on the lower level.
- The industrial systems should be easily reprogrammable frequently to support changing processes.
- System reliability and uptime is critical.

i) Regarding the data collect and store layer:

- Is there a method of collecting the data from the source?
- What processes are in use to collect data?
- Are there any technologies that could assist in the capturing, analysis or transmittal of data?
- Is there enough data storage available for the information?

j) Regarding the data contextualise layer, the following questions need to be asked:

- Is the need for information based on data analytics and or machine learning capabilities?
- Reporting and/or visualisation:
  - How will the information be presented?
  - What will the function of the information be?
- Are there any analytics capabilities that (could) detect opportunities and issues before they occur?
- Are there any big data technologies in use?
- Is there real-time visibility of the data?
- What are the evaluation capabilities of the currently implemented systems?

k) Other considerations that need special thought and planning entail the following:

- The availability of skilled technical resources to install the magnitude of sensors required in an installation that could go into the thousands of sensors.
- The cultural and organisational changes should be addressed. Also, it should be determined if is there a clear business owner that would adopt and maintain the solution.
- Personnel that use and support the technology need training for this purpose.
- The organisation needs education in terms of the value-adding possibilities of Industry 4.0.

- An industry-appropriate implementation service provider should be identified to help with implementation, should the need arise.
- What skills are needed before, during and after the IIoT implementation?

3) After the completion of the baseline assessment and the implementation design and during the execution, the guidelines for the physical implementation should consider the following:
   a) Detailed architecture is beneficial when the implementation of IIoT solutions commences.
   b) Change management during execution is needed.
   c) A skilled implementation team would ensure a high probability of success.
   d) An agile approach to implementation could be more favourable because of the rapidly changing environment.
   e) Small achievable project milestones are necessary for an IIoT implementation.

4) After the completion of the implementation of the IIoT, the guidelines after the physical implementation should consider the following:
   a) Review of the IIoT deliverables and whether they have been completed.
   b) Governance for the IIoT environment.

5.6 SUMMARY

The successful implementation of the IIoT has the potential to create value for any industrial organisation. In this chapter, the findings on the research were discussed. Given the responses, there is a need for guidelines in terms of IIoT implementation due to a high level of project planning overruns and, in some cases, the alienation of installed solutions. Regarding the people, behavioural elements play a significant role in the implementation of the IIoT.

Within the study, it was interesting to note the level of misinterpretation between the understanding and relevance of the IoT and the IIoT. All the respondents felt that the IIoT has the potential to benefit industrial organisations. Advantages include gaining market share, increased efficiencies, insight into business operations and insight into different aspects of the organisation.

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CHAPTER 6: RESEARCH CONCLUSION

6.1 INTRODUCTION

The IIoT has the potential to significantly impact the way that industrial organisations perform in terms of safety and production. Any improvements in terms of these two key performances would have an improvement in the industry. Industrial organisations are recognising this potential and are pursuing the implementation of IIoT solutions.

This study explored the literature in search of implementation guidelines and conducted interviews to extract implementation guidelines from previous implementers.

These discovered guidelines, if correctly applied, could result in the successful implementation of future IIoT implementations. A successful implementation would help realise the business value that IIoT can provide. These guidelines could also act as the starting point for IoT-specific decisions.

This chapter presents the recommendations and conclusions obtained from this specific study. It provides a set of guidelines that could aid the future implementation of IIoT

solutions. The chapter also contains a discussion on suggested recommendations for further research in this field.

6.2 RESEARCH OBJECTIVES

The research objectives were to discover a set of IIoT implementation guidelines. Industrial companies find it challenging to navigate through all the IIoT information to understand the guidelines to implement an IIoT. During the planning phase in which an organisation decides if they would want to implement an IIoT solution, these guidelines could help as a basis for an organisation to make an informed decision.

Within these objectives, the researcher also focused on extracting critical considerations for organisations that want to gain the advantages of IIoT. If the organisation understands the needed building blocks for a successful implementation, it could prevent unnecessary rework or implementation mistakes previously made by other installers and learn from these mistakes.

An understanding of lessons learned from previous installations within this industry helped to meet these objectives.

6.3 CONTRIBUTION

To ensure success regarding the implementation of the IoT in industrial environments, there are certain aspects that are not negotiable for any implementations to be a success, such as IIoT architecture. There are also recommended guidelines that would further aid in the success of implementations. This study contributes an architecture needed for the IIoT as discovered from the literature and interviews (Figure 9).

The main elements within this architecture, as described in Chapter 5, are:

- The data generation layer contains the measuring devices, also called things in the IIoT, that measure different parameters. These devices consist of centralised or decentralised hardware and software components with a high level of robustness for the continuous functioning of the hardware. Data flows from this lower level to the data collect and store layer.
- The data collect layer with the hardware and software components. The hardware could be locally on the premises or make use of offsite cloud services. The hardware is used in conjunction with software to collect and store data.
- Lastly, the data contextualise layer – be it on-site, offsite or in the cloud – presents the data in a way that the business could make sense of the data. Data becomes information within this layer and could be presented in the form of a report or a display. Analytics could also be done within this layer as well as machine learning and pattern recognition.
- A strong presence and focus on security and data integrity throughout the architecture is needed.

This study also contributes implementation guidelines within Chapter 5 that were compiled from previous learnings and implementations. These guidelines would be summarised as follows:

- There must be a clear business case or a problem to be solved by the IIoT implementation.
- The business requirement should be supported by a detailed design of the proposed IIoT implementation. The architecture of Figure 8 could be used as a basis for discussion on the components that need to be considered in compiling the detailed design. Planning for IIoT implementations must view, consider and compare the implementation in terms of the business context, centralisation or decentralisation of business, technology and the IIoT architecture. It should be kept in mind that there need to be skilled technical resources and training of unskilled resources. Also, the organisation should be educated in terms of Industry 4.0, while asking how the cultural and organisational changes will be addressed.

- After the completion of the baseline assessment and the implementation design, the execution or physical implementation takes place with a strong consideration of the change management, a skilled implementation team, an agile approach to implementation with small achievable project milestones for an IIoT implementation.

After the completion of the implementation, a review of the IIoT deliverables is needed to determine if they have been completed. The establishment of governance for the IIoT environment is also vital.

6.4 RECOMMENDATIONS

6.4.1 AWARENESS AND EDUCATION

Most of the organisations do not know what the architecture of IIoT entails or what the difference is between IoT and IIoT. The IIoT concept creates hype within organisations in the hope that it could add value. Therefore, organisations need to educate themselves on the topic to identify the differences between business cases that IIoT can solve and the sales pitches or the created hype. There is also a lack of in-house skills for IIoT implementations and the maintenance thereof after. The relevant skills at the different levels of architecture needed for IIoT implementations could pose challenges. IIoT implementations must abide by the organisations' rules and procedures.

6.4.2 COMMUNICATION

A further recommendation is that thorough discussions between network implementers and ICASA around ICASA's missing valuations of radiation patterns would be beneficial to the communications teams. Further studies on the effect of IIoT implementations within the South African regulations should be conducted.

6.4.3 TESTING

The testing of the implementation guidelines is necessary against a new IIoT implementation for effectiveness, completeness and if it enables faster implementations. The findings after testing should be made available to ensure further completeness of these guidelines.

6.4.4 CHANGE MANAGEMENT

Greater awareness around IIoT implementations should be created.

6.5 FUTURE STUDIES

The current research focused on industrial organisations within South Africa. The study offered the possibility to obtain additional value with an assessment of the IIoT challenges and guidelines in the international realm. The relative newness of the IIoT concept means that there is room for further exploration within the industrial environment.

Future studies could also revolve around the tangible (in terms of monetary value) and intangible benefits that a successful IIoT implementation could provide to organisations. Studies could also be beneficial in testing the implementation guidelines for validity against a new IIoT implementation and whether the guidelines ensure a higher success rate.

6.6 REFLECTION ON THE STUDY

This research provided implementation guidelines that industrial organisations could use for planning IIoT implementations. These guidelines could also be used to create awareness and educate organisations into the specific challenges that may arise from IIoT implementations within a South African context. Education and awareness are also needed because of the newness of the IoT in the industrial environment.

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ANNEXURE A: ETHICAL CLEARANCE RECEIVED FROM UNIVERSITY OF PRETORIA

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

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Faculty of Engineering, Built Environment and Information Technology

School of Information Technology

2019-01-29

ETHICAL CLEARANCE FOR Mr JC Jansen (Student number 18042156)

Dissertation: IMPLEMENTATION GUIDELINES FOR THE INDUSTRIAL INTERNET OF THINGS

To whom it may concern:

This is to confirm that the Research Committee of the Department of Informatics approved the application by Mr Jansen for ethical clearance. Mr Jansen complied with the standard requirements for ethical clearance as set out by the University of Pretoria's Faculty of Engineering, Built Environment and Information Technology (EBIT), as follows:

- The student signed and submitted all the application forms required for ethical clearance;
- The student submitted his/her data collection instruments for vetting by both the Research and Ethics Committees; and
- The student implemented all corrections recommended by the above-mentioned committees.

The Research Committee of the Department of Informatics therefore requests permission for Mr Jansen to collect the data needed in order to complete and submit the mini-dissertation for examination. The Committee further appreciates any effort by appropriate authorities to expedite this process, and expresses its gratitude in anticipation.

Yours sincerely

Prof Carina de Villiers
Academic Coordinator: MIT ICT Management
HOD : Informatics

ANNEXURE B: INDIVIDUAL INFORMED CONSENT FORM

Informed consent form for approval to be part of the study of:

Implementation guidelines for the Industrial Internet of Things.

I …… hereby voluntarily grant my permission for

participation in the research as explained to me by the researcher Jacobus Coenraad Jansen.

1. I am fully aware of the objective of the research, health and safety risks, and the nature of the research was fully explained.
2. I fully understand that I have a right to participate in the research and do so willingly.
3. I understand that information that is shared will be handled confidentially.
4. It has been explained to me that the research findings may be published.
5. I have received a signed copy of this form.

Signed: _____ Date: _____

Witness: _____ Date: _____

Researcher: _____ Date: _____

ANNEXURE C: COMPANIES INFORMED CONSENT FORM

Date:

To whom it may concern

I, _____, as delegated authority of _____ hereby give permission to the primary researcher Jacobus Coenraad Jansen of the School of **Informatics** at the University of Pretoria the following:

1. To engage in an interview with the employees of the above-mentioned company. I have reviewed the interview questions as per Table 1: Interview Guidelines directed at extracting implementation guidelines and key considerations for the Industrial Internet of Things, given to me by the researcher Jacobus Coenraad Jansen. I hereby give my approval for using the interview questions by the researcher.

2. To collect and publish information about the above-mentioned company that is publicly available/not available for the research project titled: Implementation Guidelines for the Industrial Internet of Things.

This authorisation is based on a mutual understanding that the above-mentioned company's name can/cannot be revealed mentioned/not mentioned anywhere in his/her project.

The information provided by the employees or any other means (such as the company's archived documents or reports) of the above-mentioned company is purely for academic purposes and cannot be used for any other purpose.

Regards,

Signature: ---------- Date:

Name & Surname:

Position/Delegation of Authority: (e.g. CEO, Director)

Tel/Email address:

ANNEXURE D: INTERVIEW TRANSCRIPTIONS

ANNEXURE D.1: INTERVIEWEE 1

Interview: IIoT implementation guidelines (responses transcript)

Respondent 1

Interview Question Category	***Sub-Category***	***Question Number***	***Interview Question***	***Interviewee Response***
Understanding the interviewee		1A	What is your primary job title?	Head of Technology for a software platform that could host IIoT
		1B	How long have you held this position?	Four months as Head of Technology and ten years' overall experience.
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Connected devices, network, on-premise and offsite services for instance cloud services.
		2B	What do you see as an industrial organisation?	Manufacturing facilities, power generation, food and beverage, water and wastewater. Organisations such as machine suppliers that would use IIoT to monitor their machines remotely.
		2C	Did you implement any IIoT solutions for industrial companies?	Implemented solutions
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	There were no formal guidelines for the implementation of the IIoT solutions.
		3B	In your view, if there are formal guidelines used what should be added to these guidelines?	No formal guidelines were used.

	Offered guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	Guidelines were based on traditional implementation methods.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Determine business value. Determine if the operations are either a greenfields operation or a brownfields operation. Conduct a baseline assessment to understand what infrastructure is in place in the brownfields operation. The IIoT should ensure safety, reliability and security. Ensure that there is a well-established network with a reputable protocol for effective communications between devices and the security of data.
Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	No data
		4B	In your opinion, what needs to be in place for an IIoT implementation?	Devices, network and cloud infrastructure
		4C	Is there any relevance if the organisation is already in existence with connected sensors or if it is a new organisation embarking on IIoT?	Brownfield operations could be more difficult to transform because they might have all seven Open Systems Interconnection (OSI) model layers implemented, and it could be costly to replace. Greenfields organisations are new implementations.
	Insights about the architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	No data
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	No data

Lessons learned		6A	Please name a given case where you have implemented an IIoT and what you would have done differently given the same case.	Implementation of a water distribution system in a rural area.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	There needs to be a safe and reliable protocol.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	No data
		6D	What would you describe as the advantages of an IIoT implementation?	No data

ANNEXURE D.2: INTERVIEWEE 2

Interview: IIoT implementation guidelines (responses transcript)

Respondent 2

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Executive: Solutions and innovations for an IIoT company that provide network communication.
		1B	How long have you held this position?	Two years.
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Companies that want to increase efficiencies in terms of manufacturing, supply chain and cost savings.
		2B	What do you see as an industrial organisation?	Organisations that produce something.
		2C	Did you implement any IIoT solutions for industrial companies?	I assist our channel partners with IIoT solutions for implementation.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	Formal guidelines for communications are ICASA (Independent Communication Association of South Africa) licencing.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	ICASA is missing valuations of radiation patterns.

	Offered Guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	No data
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	The industrial protocols need to be adapted to the low-power wide area network technology.
Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	Any industry that manufactures things.
		4B	In your opinion, what needs to be in place for an IIoT implementation?	The right partner with the appropriate experience in delivering IIoT solutions in the specific industry. Infrastructure and network coverage.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	No data
	Insights about the architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	Sensors at the bottom layer in line of what needs to be measured, for example pressure, temperature. An internal or external network. An edge device or cloud platform. A storage device to store the data. Using cloud services for analytics. Also, ways to visualise it to make some intelligent business decision. Drawing correlations between different data sets and data points across multiple locations. Sensors at the bottom layer in line with what needs to be measured, for example pressure, temperature. An internal or external network. An edge device or cloud platform. A storage device to store the data. Using cloud services for analytics. Also, ways to visualise it to make some intelligent business decision. Drawing

				correlations between different data sets and data points across multiple locations to make informed decisions for business decisions.
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	What is the cost of the problem to be solved? Is it a problem? How big is the problem? Is technology the solution? Define the business case properly upfront.
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case.	Water meter installations ensure that the hardware is reliable. The business case depends on the sensor being available for an x amount of time.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	As per 6A
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT.	Pick a problem that exists, test it with an existing solution. The advantage of existing solution is that they could easily and quickly be implemented to prove value. Have the budget upfront and understand the timelines for the solution. Also, who will provide it?
		6D	What would you describe as the advantages of an IIoT implementation?	Gaining market share and increase efficiencies. Could pass on savings to your customers or offer additional services, insight into your business operations, giving customers insight into different stages within the business.

ANNEXURE D.3: INTERVIEWEE 3

Interview: IIoT Implementation Guidelines (Responses Transcript)

Respondent 3

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Solutions Architect
		1B	How long have you held this position?	Thirteen years.
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Connected things in the manufacturing aspects of the business that help with the digitisation of the business. The aim to create even more useful data than the past.
		2B	What do you see as industrial organisations?	Involves manufacturing or converting raw material into something that is an end product.
		2C	Did you implement any IIoT solutions for industrial companies?	No, but I design for implementation teams and advise them.
	Existing guidelines	3A	Are there any formal guidelines used in implementing the IIoT in the industrial organisation?	There are guidelines out there.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	There is a people aspect missing in terms of change management. The end-user does not use implemented solutions.

	Offered guidelines	3C	Are there any informal guidelines that used in implementing the IIoT in the industrial organisation?	Firstly, assess the current business environment. Compare this environment against where the business would want to position themselves in the future. They also need to be educated into what the possibilities are in terms of Industry 4.0. There needs to be a clear guideline in terms of what the business information needs are. The storage that you would need to host the data.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Be aware of constant changes in the IIoT environment; for example, more reliable, intelligent sensors and protocols are continuously created.
Key considerations before the implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	All of the industries could benefit from it.
		4B	In your opinion, what needs to be in place for an IIoT implementation?	The people aspect of the adoption of the IIoT. The right connectivity, protocols, and technology to talk to different devices. The devices should also be able to handle the environments in which it functions. If there is no connectivity to the cloud, handle analytics on-site rather than putting the business at risk via an unstable internet link.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	A new organisation would have no challenges in terms of legacy systems.
	Insights about the Architecture of IIoT	5A	Please draw the architecture needed for IIoT implementation.	At the bottom layer, the sensors with communications that relay the information. The communications layer could be wired or wireless with different protocols. Followed by a database layer that would host the data. On top of that is the analytics layer that could draw correlations between various database sources

		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	Manage the organisational change component, and the people change aspect.
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same situation.	Without the proper training and business adoption, any implemented technology intended for the better of the organisation might not even be used.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	Previously answered.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT	Define the end state of where you want to be with the solution and follow an agile approach to reaching this stage because it is a changing environment.
		6D	What would you describe as the advantages of an IIoT implementation?	Tracking a lot more data for more informed decisions and higher business value could be obtained by liaising with other companies from the outside. Sensors are becoming cheaper.

ANNEXURE D.4: INTERVIEWEE 4

Interview: IIoT implementation guidelines (responses transcript)

Respondent 4

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Co-founder and Director of an IIoT implementations company.
		1B	How long have you held this position?	Seventeen months
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Using devices connecting to networks on and then taking that data and analysing it to make sure that peoples jobs more efficient.
		2B	What do you see as industrial organisations?	Mining, manufacturing, food and beverage
		2C	Did you implement any IIoT solutions for industrial companies?	Yes
	Existing guidelines	3A	Are there any formal guidelines used in implementing the IIoT in the industrial organisation?	No formal guidelines; we make use of our own set of guidelines.
		3B	In your view, if there are formal guidelines used what should be added to these guidelines?	No formal guidelines.

	Offered guidelines	3C	Are there any informal guidelines that used in implementing the IIoT in the industrial organisation?	Know the requirements of the customer and compile a user requirement document. Create a design document.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Establish how you would acquire a reliable and consistent flow of data (communication) via a network. Establish if there is a network available for that reason. Identify the platform needed to analyse and interpret the data. This platform will need to integrate into existing organisational platforms.
Key considerations before the implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	Any industry from mining, food and beverage, manufacturing, retail could benefit from an IIoT solution.
		4B	In your opinion, what needs to be in place for an IIoT implementation?	Devices, networks and a platform is one of the prerequisites.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	Yes, a greenfields organisation makes it easier than a brownfields because the IIoT implementer could decide on the various infrastructure and ensure compatibility. A brownfields organisation would require more development hours for integration between existing and new.
	Insights about the architecture of IIoT	5A	Please draw the architecture needed for IIoT implementation.	On the field level, there is sensors and devices linked to a network. This network connects data acquired from the to the cloud to enable analytics.
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	There should be a problem that needs to be solved that would create business value and a return on investment.

Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same situation.	An implementation was done at a healthcare facility to monitor fridge temperature for the cooling of pharmaceuticals. It is imperative to fully understand the customer's expectations and document this as formal requirements. There wasn't a reliable network connection.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	To methodically lay out the whole process of where the data originates. Where the data should go and what the purpose of the data is together with the need for it before any implementation starts. Implementation at a large scale would mean a few months to install the necessary units — the maintenance of the whole IIoT solution after the time is also a consideration.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	Liaising with a competent service provider to implement successful solutions.
		6D	What would you describe as the advantages of an IIoT implementation?	Would describe the benefits as vast and exponential.

ANNEXURE D.5: INTERVIEWEE 5

Interview: IIoT implementation guidelines (responses transcript)

Respondent 5

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Senior Project Manager in the field of information management.
		1B	How long have you held this position?	More than 28 years.
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Using modern media, modern technology in the industrial environment to streamline processes to improve efficiencies, control plant better, which will go from your plant to your phone at home, so completely integrated solutions.
		2B	What do you see as industrial organisations?	Every organisation that's part of manufacturing or mining or all the big industries, food, pharmaceuticals.
		2C	Did you implement any IIoT solutions for industrial companies?	I am busy implementing several implementations for the organisation that I am employed in at the moment.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	There were no guidelines; I think we are learning as we go along. Only from a project management view, we use formal guidelines of project management according to PM BOK.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	No formal guidelines in terms of IIoT

	Offered Guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	New equipment is introduced into the industrial organisations, for instance tablets and phones. It is important to establish the connectivity to these new devices, and the capabilities of these devices before any implementation could happen. Different platforms, for instance Windows and Android create security challenges.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Determine hardware compatibility for the proposed solution. The security of the hardware. Clearly define your to be achieved. These KPIs should be tracked and managed throughout the Project.
Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	All industries
		4B	In your opinion, what needs to be in place for an IIoT implementation?	The architecture of the solution needs to be detailed and not high level
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	IIoT can complement existing investments in technology.
	Insights about the Architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	The respondent felt that an architect could better answer this and that his focus is on the implementation of the solution.
		5B	In your opinion, what is some of the key considerations for any industrial organisation embarking on an IIoT implementation?	The cost of the solution, a strong team that has the knowledge and training for users.
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case.	The digitalisation of the pre-use inspection of equipment in mining on tablets.

		6B	What would you describe as the lessons learned from the specific case in regards to an IIoT implementation?	Lack of detailed design in terms of IIoT architecture. Hardware compatibility. Some designs were based on technology and not fit for purpose.
		6C	What advicee could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	As mentioned in 6B
		6D	What would you describe as the advantages of an IIoT implementation?	Improved efficiency of the process, production improvements.

ANNEXURE D.6: INTERVIEWEE 6

Interview: IIoT implementation guidelines (responses transcript)

Respondent 6

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Solution Business Analyst in automation
		1B	How long have you held this position?	Six years of experience in the field
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Connecting anything anywhere and obtaining and moving data to a centralised place for analysis and visualisation of the data.
		2B	What do you see as industrial organisations?	Scalability and size of the organisation, e.g. mining.
		2C	Did you implement any IIoT solutions for industrial companies?	Radio communications on decentralised mobile substations that collect data on the substations.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	Site-specific designs in terms of hardware.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	Change management is needed.
	Offered guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	Nothing to mention.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Nothing to mention.

Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	Remote areas, for example, farming.
		4B	In your opinion, what needs to be in place for an IIoT implementation?	Peoples skill and business requirements.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	If there is existing infrastructure, the ability to use the same infrastructure for new implementations.
	Insights about the Architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	A platform for the hosting of the data. The data layer could also be in the cloud. Network connectivity and the measuring device.
		5B	In your opinion, what is some of the key considerations for any industrial organisation embarking on an IIoT implementation?	Is there value in the implementation and do you know what the outcome should be?
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case.	Challenges were the people skills in terms of the IIoT solution and the compatibility of the existing infrastructure. The protocols in use. The legacy devices that were not able to help with IoT.
		6B	What would you describe as the lessons learned from the specific case in regards to an IIoT implementation?	Nothing additional to add.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT	Ensure the business need.
		6D	What would you describe as the advantages of an IIoT implementation?	Connecting remote devices.

ANNEXURE D.7: INTERVIEWEE 7

Interview: IIoT implementation guidelines (responses transcript)

Respondent 7

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Electrical Engineering Technologist in the field of automation.
		1B	How long have you held this position?	Twenty years of experience.
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Separate devices are interacting with each other and integrating with internet that host cloud services.
		2B	What do you see as industrial organisations?	Companies
		2C	Did you implement any IIoT solutions for industrial companies?	Hand-held communication devices, energy consumption information and statistics needed in a central location.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	No, because it is a new field.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	Nothing to add.
	Offered Guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	Network reliability and stability. Reliability of measuring devices in the harsh mining environment.

		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	As per question 3C
Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	All of the industries
		4B	In your opinion, what needs to be in place for an IIoT implementation?	Robust IIoT devices and good network connectivity (including redundancy should there be a failure) with a high level of security
		4C	Are there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	Respondent's answer is not to the point.
	Insights about the architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	Devices, network and security.
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	Security.
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case?	Hand-held communication devices, energy consumption information and statistics needed in a central location.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	Robust IIoT devices are needed.

		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	Ensure the ruggedness of the IIoT devices.
		6D	What would you describe as the advantages of an IIoT implementation?	Quick business access from a central location for business information decisions.

ANNEXURE D.8: INTERVIEWEE 8

Interview: IIoT implementation guidelines (responses transcript)

Respondent 8

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Information Management Project Manager
		1B	How long have you held this position?	Twenty years
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	IIoT is around making the industry smart, making the industry better, what can be done to make a specific industry more efficient, cause everybody wants to be more efficient so that they can, at the end of the day, make more money.
		2B	What do you see as industrial organisations?	Farming and manufacturing, for instance coal.
		2C	Did you implement any IIoT solutions for industrial companies?	Was involved in the ID cards in South Africa and the scanners.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	No specific guidelines.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	No specific guidelines.

	Offered guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	Experience used as a guideline, a need for the right skills.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Nothing to add.
Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	All industries
		4B	In your opinion, what needs to be in place for an IIoT implementation?	People with the relevant knowledge, understanding of the business requirements, a vision of what is needed, the technology.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	Uncertain
	Insights about the Architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	Measuring equipment that gathers the data for information, connected to a physical network for communication to data storage or cloud services.
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	Industrial strength equipment, strong communication infrastructure. The right application for the data
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case.	The ID card implementation

		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	Breakdown deliverables into smaller achievable steps.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	Successful IIoT implementation could mean that you might be able to produce more. Do you have a market for more product? Is logistics in place?
		6D	What would you describe as the advantages of an IIoT implementation?	Efficiency

ANNEXURE D.9: INTERVIEWEE 9

Interview: IIoT implementation guidelines (responses transcript)

Respondent 9

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Solutions Architect: anything to do with IoT and connectivity
		1B	How long have you held this position?	In this position: 9 months
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Devices talking directly to cloud platforms and the like.
		2B	What do you see as industrial organisations?	Mines, industry, manufacturing plants and FMCG
		2C	Did you implement any IIoT solutions for industrial companies?	A monitoring system in a food manufacturing plant.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	There are not any guidelines in place except for manufacture recommended specifications for the products themselves.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	Because of no formal guidelines.
	Offered guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	Important to abide by the company's non-disclosure agreements. Site-specific constraints for the installation of sensors. As-is assessment.

		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	Understand the technology that you want to implement and the environment that you want to improve. The cost involved.
Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	All industries
		4B	In your opinion, what needs to be in place for an IIoT implementation?	Relevant procedures, the correct technology (right sensors in the right environment) and network.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	It is possible to accommodate the existing infrastructure.
	Insights about the Architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	Sensors at the edge of the field, the gateway to the cloud and processing in the cloud. In the cloud platform, you have dashboards and analytics.
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	Type of technology used. Uptime of the system is critical.
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case.	Previously answered.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	Previously answered.

		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	Do not just implement the first option that you see, do the research first.
		6D	What would you describe as the advantages of an IIoT implementation?	The interconnectivity of everything: insight into things that you did not previously have.

ANNEXURE D.10: INTERVIEWEE 10

Interview: IIoT implementation guidelines (responses transcript)

Respondent 10

Interview Question Category	*Sub-Category*	*Question Number*	*Interview Question*	*Interviewee Response*
Understanding the interviewee		1A	What is your primary job title?	Senior Technical architect
		1B	How long have you held this position?	Two years, with 19 years' overall experience
Insight about IIoT practice	Defining the landscape	2A	How would you identify the IIoT?	Connected things that could generate data.
		2B	What do you see as industrial organisations?	Mining
		2C	Did you implement any IIoT solutions for industrial companies?	I was involved from an architectural perspective.
	Existing guidelines	3A	Are there any formal guidelines that were used in implementing the IIoT in the industrial organisation?	No uniform guidelines.
		3B	In your view, if there are formal guidelines used, what should be added to these guidelines?	No uniform guidelines.
	Offered guidelines	3C	Are there any informal guidelines that were used in implementing the IIoT in the industrial organisation?	Guidelines based on what the client needs.
		3D	What guidelines for an IIoT would you suggest would aid future IIoT implementations?	No relevant input.

Key considerations before implementation of IIoT	Insights about the industries for IIoT	4A	In your view, what industries could benefit IIoT implementations?	Every industry
		4B	In your opinion, what needs to be in place for an IIoT implementation?	Proper governance. Structured and unstructured data. An as-is assessment of the current state of your data.
		4C	Is there any relevance if the organisation is in existence already with connected sensors or if it is a new organisation embarking on IIoT?	No meaningful answer received.
	Insights about the architecture of IIoT	5A	Please draw the architecture needed for an IIoT implementation.	Data collection zone to pull in or push the needed information. Data storage zone for the storage of structured and unstructured data.
		5B	In your opinion, what are some of the key considerations for any industrial organisation embarking on an IIoT implementation?	What need is there to be fulfilled?
Lessons learned		6A	Please name a given case where you implemented an IIoT and what you would have done differently given the same case.	In real-time, an indication of what your network is doing, if any ports are down if any switches need replacing.
		6B	What would you describe as the lessons learned from the specific case in regard to an IIoT implementation?	What the client's needs are.
		6C	What advice could you give to any future industrial organisation that would want to implement the IIoT to gain advantages of IIoT?	Start small.
		6D	What would you describe as the advantages of an IIoT implementation?	The advantage in terms of monetary value.